## Preface

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# Chapter 1
## Normed Linear Spaces

### § 1.1. Basic Definitions and Results

**Definition 1.1.1.** A *seminorm* on an $\mathbb{F}$-vector space $X$ is a map:

$$\|\cdot\| : X \longrightarrow [0, \infty)$$

satisfying:

(1) $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{F}$ and $x \in X$ (*homogeneity*);

(2) $\|x + y\| \leqslant \|x\| + \|y\|$ for all $x, y \in X$ (*triangle inequality*).

If the following condition is also satisfied:

(3) $\|x\| \geqslant 0$ and $\|x\| = 0$ if and only if $x = 0$ (*positive definiteness*),

then we say $\|\cdot\|$ is a *norm*. The ordered pair $(X, \|\cdot\|)$ is called a *normed linear space*.

*Remark.* In most cases, one can take $\mathbb{F}$ to be $\mathbb{R}$ or $\mathbb{C}$ without any loss of generality.

*Remark.* A normed linear space induces a metric by the equation $d(x, y) := \|y - x\|$.

*Remark.* A normed linear space is finite dimensional if it is as a vector space; otherwise it is infinite dimensional.

**Definition 1.1.2.** Let $(X, \|\cdot\|)$ be a normed linear space.

(1) The *open ball centered around* $x_0 \in X$ *with radius* $r > 0$ is denoted
$B(x_0, r) := \{x \in X \mid \|x - x_0\| < r\}$.

(2) The *closed ball centered around* $x_0 \in X$ *with radius* $r > 0$ is denoted
$\overline{B}(x_0, r) := \{x \in X \mid \|x - x_0\| < r\}$.

(3) The *sphere centered around* $x_0 \in X$ *with radius* $r > 0$ is denoted
$S(x_0, r) := \{x \in X \mid \|x - x_0\| = r\}$.

**Definition 1.1.3.** Two norms $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ on a normed linear space $X$ are said to be *equivalent* if there exists a constant $C \geqslant 1$ such that, for all $x \in X$:

$$\frac{1}{C} \|x\|_\beta \leqslant \|x\|_\alpha \leqslant C \|x\|_\beta .$$

**Example 1.1.1.** Let $x \in \mathbb{R}^n$, where $x := (x_1, ., , x_n)$ Then:

$$\|x\|_2 := \left(\sum_{i=1}^{n} |x_i|^2\right)^{\frac{1}{2}},$$

$$\|x\|_1 := \sum_{i=1}^{n} |x_i|,$$

$$\|x\|_\infty := \max_{i=1}^{n} |x_i|^p,$$

$$\|x\|_p := \left(\sum_{i=1}^{n} |x_i|^p\right)^{\frac{1}{p}}, \; p > 1$$

are norms on $\mathbb{R}^n$.

**Example 1.1.2.** Let $a, b \in \mathbb{R}$ with $a < b$. The set of continuous real-valued functions on $[a, b]$ is denoted $C([a, b]) := \{f : [a, b] \to \mathbb{R} \mid f \text{ is continuous }\}$. Then:

$$\|f\|_\infty := \sup\{|f(x)| \mid x \in [a, b]\},$$

$$\|f\|_2 := \left(\int_a^b |f(x)|^2 dx\right)^{\frac{1}{2}},$$

$$\|f\|_p := \left(\int_a^b |f(x)|^p dx\right)^{\frac{1}{p}},$$

are all norms on $C([a, b])$.

**Definition 1.1.4.** Let $(X, \|\cdot\|)$ be a normed linear space. A subset $V \subset X$ is <u>*convex*</u> if for all $x, y \in V$ and for all $t \in [0, 1]$, $ty + (1 - t)x \in V$.

**Example 1.1.3.** Any subspace of a normed linear space is convex. Moreover, by the triangle inequality any open ball is convex.

*Remark.* Note that the *supremum norm* $\|\cdot\|_\infty$ on $C([a, b])$ only exists (by the Extreme Value Theorem) because $[a, b]$ is compact. Hence the above example can be generalized to a compact metric space $Y$. In particular, for $C(Y) := \{f : Y \to \mathbb{R} \mid f \text{ is continuous }\}$, the norm $\|f\|_\infty := \sup\{|f(x)| \mid x \in Y\}$ always exists and is finite.

**Proposition 1.1.1.** *Let $(X, d)$ and $(Y, \rho)$ be metric spaces. Let $f : X \to Y$. Then $f$ is continuous if and only if the preimage of an open set in $Y$ is open in $X$.*

*Proof.* Suppose $f$ is continuous. Let $V \subseteq Y$ be open. If $f^{-1}(V) = \emptyset$, then there is nothing to show. If not, pick any $x \in f^{-1}(V)$. Then $f(x) \in V$. Since $V$ is open, there exists $\epsilon > 0$ such that $f(x) \in B(f(x), \delta) \subseteq V$. Since $f$ is continuous, there exists a $\delta > 0$ such that $f(B(x, \delta)) \subseteq B(f(x), \epsilon) \subseteq V$. Thus $B(x, \delta) \subseteq f^{-1}(V)$; i.e., $f^{-1}(V)$ is open.

Conversely, let $x \in X$ and $\epsilon > 0$. Since $B(f(x), \epsilon)$ is open in $X$, $f^{-1}(B(f(x), \epsilon))$ is open in $Y$. Clearly $x \in f^{-1}(B(f(x), \epsilon))$. Since this set is open, there exists $\delta > 0$ such that $x \in B(x, \delta) \subseteq f^{-1}(B(f(x), \epsilon))$. Hence $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$. Since the set of all open balls is a basis for a topology, any open set can be written as the union of open balls. Thus $f$ is continuous. □

*Remark.* It follows that $f$ is continuous if and only if the preimage of a closed set is closed.

**Definition 1.1.5.** Let $(X, \|\cdot\|)$ be a normed linear space and $\{x_n\}_{n=1}^{\infty}$ be a sequence in $X$.

(1) The sequence $\{x_n\}_{n=1}^{\infty}$ *converges to $x \in X$ in the $\|\cdot\|$-norm* if $\|x - x_n\| \to 0$.

(2) A *series in $X$* is a formal sum $\sum_{n=1}^{\infty} x_n$. We say the series $\sum_{n=1}^{\infty} x_n$ *converges to $x \in X$* if the sequence of partial sums $\left\{\sum_{i=1}^{n} x_i\right\}_{n=1}^{\infty}$ converges to $x$. We say $\sum_{n=1}^{\infty}$ is *absolutely convergent* if $\sum_{n=1}^{\infty} \|x_n\|$ converges.

**Proposition 1.1.2.** *Let $X$ be a normed linear space. The following are equivalent:*

*(1) $X$ is a Banach space;*

*(2) Every absolutely convergent series in $X$ converges.*

*Proof.* □

## § 1.2. Completeness and Compactness

**Lemma 1.2.1.** *In $(\mathbb{R}^n, \|\cdot\|_\infty)$, every closed ball $\overline{B}(0, c)$ is compact.*

*Proof.* □

**Theorem 1.2.1.** *Let $X$ be a finite dimensional normed linear space. Every closed and bounded subset of $X$ is compact.*

*Proof.* □

**Corollary 1.2.1.** *Every finite dimensional normed linear space is Banach.*

*Proof.* Let $X$ be a finite dimensional normed linear space and $\{x_n\}_{n=1}^{\infty}$ a Cauchy sequence. Then $\{x_n\}_{n=1}^{\infty}$ is closed and bounded, hence it is compact. Thus $\{x_n\}_{n=1}^{\infty}$ admits a convergent subsequence. But since $\{x_n\}_{n=1}^{\infty}$ is a Cauchy sequence which admits a convergent subsequence, $\{x_n\}_{n=1}^{\infty}$ is convergent. Thus $X$ is Banach. □

**Corollary 1.2.2.** *Every finite dimensional subspace of a normed linear space is closed.*

*Proof.* This follows from Corollary 1.2.1 and Exercise 1.2.4. □

**Lemma 1.2.2** (Riesz's Lemma)**.** *Let* $(X, \|\cdot\|)$ *be a normed linear space. Let* $Z$ *be a closed and proper subspace of* $X$*. Let* $0 < \lambda < 1$ *be fixed. Then there exists some* $x \in X \setminus Z$ *such that* $\|x\| = 1$ *and* $\operatorname{dist}(x, Z) > 1$.

*Proof.* □

**Theorem 1.2.2.** *Let* $X$ *be a normed linear space. Then* $X$ *is finite dimensional if and only if* $\overline{B}(0, 1)$ *is compact.*

*Proof.* □

## Exercises

**Exercise 1.2.1.** *Let* $K$ *be a compact metric space (fine to assume that* $K = [0, 1]$*). Show that* $C(K)$ *paired with the the sup-norm* $\|\cdot\|_\infty$ *is Banach.*

*Proof.* □

**Exercise 1.2.2.** *In any metric space, a closed subset of a compact set is compact.*

*Proof.* □

**Exercise 1.2.3.** *In any metric space, a Cauchy sequence which has a convergent subsequence must converge.*

*Proof.* □

**Exercise 1.2.4.** *A subspace of a Banach space is closed if and only if it is Banach.*

*Proof.* Let X be a Banach space and Y a subspace. Suppose that Y is closed. Let $\{y_n\}_{n=1}^\infty \subseteq Y$ be a Cauchy sequence. Then $\{y_n\}_{n=1}^\infty$ is Cauchy in $X$. Therefore $y_n \to x$ for some $x \in X$. But since $Y$ is closed, it contains all of its limit points. Hence $x \in Y$. Therefore $\{y_n\}_{n=1}^\infty$ converges in Y, establishing that Y is Banach.

Conversely, suppose that Y is Banach. Let $\{y_n\}_{n=1}^\infty \subseteq Y$ be a sequence such that $y_n \to x \in X$. Note that $\{y_n\}_{n=1}^\infty$ is convergent, hence Cauchy. But since $Y$ is Banach, $\{y_n\}_{n=1}^\infty$ converges to some $y \in Y$. It must be the case that $x = y \in Y$. Therefore Y contains all of its limit points; i.e., it is closed in X. □

**Exercise 1.2.5.** *In any normed linear space $X$ with closed subspace $Z$, if $x \in X \backslash Z$ then $\operatorname{dist}(x, Z) > 0$.*

*Proof.* Suppose towards contradiction $\operatorname{dist}(x, Z) = \inf\{\|x - z\| \mid z \in Z\} = 0$. We can find a sequence $\{z_n\}_{n=1}^{\infty} \subseteq Z$ such that $\|x - z_n\| \longrightarrow 0$. Hence $z_n \longrightarrow x$. But since $Z$ is closed, $x \in Z$. This is a contradiction, hence $\operatorname{dist}(x, Z) > 0$. □

**Exercise 1.2.6.** *In any normed linear space $X$ with subspace $Z$, if $x \in X$ and $z \in Z$ then $\operatorname{dist}(x, Z) = \operatorname{dist}(x + z, Z)$.*

*Proof.* Note that $\operatorname{dist}(x+z, Z) = \inf\{\|(x+z) - z_0\| \mid z_0 \in Z\} = \inf\{\|x - (z_0 - z)\| \mid z_0 \in Z\}$. Since $\{z_0 - z \mid z_0 \in Z\} = Z$, we have that $\inf\{\|x - z\| \mid z \in Z\} = \inf\{\|(x+z) - z_0\| \mid z_0 \in Z\}$; i.e., $\operatorname{dist}(x, Z) = \operatorname{dist}(x+z, Z)$. □

**Exercise 1.2.7.** *Let $X$ and $Y$ be normed linear spaces and let $\varphi : X \longrightarrow Y$ be a linear operator. Show that $\varphi$ is continuous if and only if it is continuous at 0.*

*Proof.* □

## § 1.3. Linear Operators

**Definition 1.3.1.** Let $X$ and $Y$ be normed linear spaces. A map $T : X \longrightarrow Y$ is linear if $T(x + \alpha y) = T(x) + \alpha T(y)$ for all $x, y \in X$ and $\alpha \in \mathbb{F}$. $T$ is called a *linear transformation* (or *linear operator*). If $X$ is a normed linear space over $\mathbb{F}$, then a linear transformation $\varphi : X \longrightarrow \mathbb{F}$ is called a *linear functional*.

*Remark.* The norm on $\mathbb{F}$ is the usual absolute value in $\mathbb{R}$ or modulus operator in $\mathbb{C}$.

**Example 1.3.1.** Linear operators between $\mathbb{R}^n$ and $\mathbb{R}^m$ correspond to $n \times m$ matrices.

**Example 1.3.2.** Consider the normed linear space $\left(C([0,1]), \|\cdot\|_\infty\right)$. Evaluation at zero is a linear functional, as well as the map $x \mapsto \int_0^1 x(t)dt$. Moreover, $T : C([0,1]) \longrightarrow C([0,1])$ defined by $x(t) \mapsto \int_0^1 K(s,t)x(s)ds$ is a linear transformation, where $K$ is a continuous function on $[0,1] \times [0,1]$.

**Definition 1.3.2.** Let $X$ and $Y$ be normed linear spaces and $T : X \longrightarrow Y$ a linear transformation. $T$ is *bounded* if there exists $M > 0$ such that for all $x \in X$, $\|x\| \leqslant 1$ implies $\|Tx\| \leqslant M$.

**Theorem 1.3.1.** *Let $X$ and $Y$ be normed linear spaces. A linear transformation $T : X \longrightarrow Y$ is bounded if and only if it is continuous.*

*Proof.* Suppose $T$ is continuous. Then $T$ is continuous at zero. Choose $\delta > 0$ such that $\|x\| < \delta$ implies $\|Tx\| < 1$. Now let $x \in X$ with $\|x\| \leqslant 1$. Then $\|Tx\| = \left\|T\left(\frac{x\delta}{\delta}\right)\right\| = |\frac{1}{\delta}|\,\|T(\delta x)\| \leqslant \frac{1}{\delta}$. Therefore $T$ is bounded.

Conversely, let $M > 0$ satisfy $\|Tx\| \leqslant M$ for all $x \in \overline{B}(0,1)$. Let $\epsilon > 0$. Choose $\delta = \frac{\epsilon}{M}$. Then $\|x\| \leqslant \delta$ implies $\left\|\frac{M}{\epsilon}x\right\| \leqslant 1$. Hence by our assumption $\left\|T\left(\frac{M}{\epsilon}x\right)\right\| \leqslant M$, which is equivalent to $\|Tx\| \leqslant \epsilon$. Thus $T$ is continuous at $0$, hence it is continuous everywhere. □

**Example 1.3.3.** Consider $D : C^1([0,1]) \to C([0,1])$ given by $Df = f'$. Consider the sequence of functions $\{f_n\}_{n=1}^{\infty} \subseteq C^1([0,1])$ given by $f_n(x) = \frac{1}{\sqrt{n}}x^n$. Then for all $n \in \mathbb{N}$, $Df_n(x) = \sqrt{n}x^{n-1}$. Hence $D$ is unbounded.

**Theorem 1.3.2.** *Let $X$ be a normed linear space. A linear functional $\varphi : X \to \mathbb{F}$ is continuous if and only if its kernel is closed.*

*Proof.* Suppose $\varphi$ is continuous. Then $\ker\varphi = \varphi^{-1}(\{0\})$ is closed, as the continuous preimage of a closed set is closed.

Conversely, suppose towards contradiction $\varphi$ is not continuous. Then $\varphi$ is unbounded; i.e., there exists a sequence $\{x_n\}_{n=1}^{\infty} \subset X$ such that $\|x_n\| = 1$ for all $n \in \mathbb{N}$ and $|\varphi(x_n)| \to \infty$. If $\ker\varphi = X$ then there is nothing to show. Assume $\ker\varphi \neq X$ and pick some $x \in X \setminus \ker\varphi$. Define $y_n := x - \frac{\varphi(x)}{\varphi(x_n)}x_n$. Then $\varphi(y_n) = \varphi(x) - \frac{\varphi(x)}{\varphi(x_n)}\varphi(x_n) = 0$. Hence $y_n \in K$ for all $n \in \mathbb{N}$. But note that $\|y_n - x\| = \left\|-\frac{\varphi(x)}{\varphi(x_n)}x_n\right\| = \frac{|\varphi(x)|}{|\varphi(x_n)|}\|x_n\| \to 0$. Hence $y_n \to x$. Since $\ker\varphi$ is closed, $x \in \ker\varphi$. This is a contradiction, establishing the proof. □

**Theorem 1.3.3.** *Let $X$ be a finite dimensional normed linear space and $Y$ a normed linear space. If $T : X \to Y$ is linear, then $T$ is continuous.*

*Proof.* Pick a basis for $X$ such that $X = \text{span}\{b_1, ..., b_n\}$. Define $M := \max_{i=1}^{n}\{T(b_i)\}$. For each $j = 1, ..., n$, define a linear functional $\varphi_j : X \to \mathbb{F}$ by:

$$\begin{aligned}\varphi_j(x) &= \varphi_j\left(\sum_{i=1}^{n}\alpha_i b_i\right)\\ &= \sum_{i=1}^{n}\alpha_i\varphi_j(b_i)\\ &:= \alpha_j\end{aligned}$$

Since all subspaces of $X$ are closed by Corollary 1.2.2, surely $\ker\varphi_j$ is closed. Hence by Theorem 1.3.2 $\varphi_j$ is continuous. Since $\varphi_j$ is continuous, find $L > 0$ sufficiently large enough such that for every $x \in X$ with $\|x\| \leqslant 1$, we have $|\varphi_j(x)| \leqslant L$. Since $T : X \to Y$ is linear:

$$\begin{aligned}
\|Tx\| &= \left\| T \sum_{i=1}^{n} \alpha_i b_i \right\| \\
&= \left\| \sum_{i=1}^{n} \alpha_i T b_i \right\| \\
&= \left\| \sum_{i=1}^{n} \varphi_i(x) T b_i \right\| \\
&\leqslant \sum_{i=1}^{n} |\varphi_i(x)| \, \|Tb_i\| \\
&\leqslant nLM \\
&< \infty.
\end{aligned}$$

Thus T is continuous. □

**Theorem 1.3.4.** *Let* $X$ *be a finite dimensional vector space. All norms on* $X$ *are equivalent.*

*Proof.* Let $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ be two norms. Consider $\mathrm{id} : (X, \|\cdot\|_\alpha) \to (X, \|\cdot\|_\beta)$. This map is surely linear, hence by Theorem 1.3.3 id is continuous. Hence we can find some $A > 0$ sufficiently large such that, for all $x \in X$ with $\|x\|_\alpha \leqslant 1$, we have $\|x\|_\beta \leqslant A$. Thus:

$$\begin{aligned}
\|x\|_\beta &= \left\| \|x\|_\alpha \frac{x}{\|x\|_\alpha} \right\|_\beta \\
&= \|x\|_\alpha \left\| \frac{x}{\|x\|_\alpha} \right\|_\beta \\
&\leqslant \|x\|_\alpha A.
\end{aligned}$$

Applying the same argument to $\mathrm{id}^{-1}$ gives $\|x\|_\alpha \leqslant \|x\|_\beta B$, establishing the theorem. □

## § 1.4. Bounded Linear Functionals

**Definition 1.4.1.** Let X and Y be normed linear spaces.

(1) The space of bounded linear operators between X and Y is denoted $\mathcal{L}(X, Y) := \{T : X \to Y \mid T \text{ is bounded}\}$.
(2) The *operator norm* of a bounded linear operator $T \in \mathcal{L}(X, Y)$ is $\|T\|_{\mathcal{L}(X,Y)} := \sup\{\|Tx\|_Y \mid x \in \overline{B}_X(0, 1)\}$.
(3) The *continuous dual of* X is defined as $X^* := \mathcal{L}(X, \mathbb{F})$.

*Remark.* We denote $\mathcal{L}(X, X)$ by $\mathcal{L}(X)$.

**Proposition 1.4.1.** *If* $T \in \mathcal{L}(X, Y)$ *and* $x \in X$, *then* $\|Tx\| \leqslant \|T\|_{\mathcal{L}(X,Y)} \cdot \|x\|$.

*Proof.* If $x = 0$, then there is nothing to show. If $x \neq 0$, then:

$$\begin{aligned} \|Tx\| &= \left\| \|x\| \, T\left(\frac{x}{\|x\|}\right) \right\| \\ &= \left\| T\left(\frac{x}{\|x\|}\right) \right\| \|x\| \\ &\leqslant \|T\|_{\mathcal{L}(X,Y)} \|x\| . \end{aligned}$$

□

**Proposition 1.4.2.** *If* $S, T \in \mathcal{L}(X)$, *then* $S \circ T \in \mathcal{L}(X)$.

*Proof.* Let $x \in X$ such that $\|x\| \leqslant 1$. By Proposition 1.4.1:

$$\begin{aligned} \|(S \circ T)(x)\| &\leqslant \|S(T(x))\| \\ &\leqslant \|S\|_{\mathcal{L}(X)} \|Tx\| \\ &\leqslant \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)} \|x\| \\ &\leqslant \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)} . \end{aligned}$$

Taking the supremum over all $\|x\| \leqslant 1$ gives $\|S \circ T\|_{\mathcal{L}(X)} \leqslant \|S\|_{\mathcal{L}(X)} \|T\|_{\mathcal{L}(X)}$. □

**Theorem 1.4.1.** *Let* $X$ *be a normed linear space and* $Y$ *a Banach space. Then* $(\mathcal{L}(X, Y), \|\cdot\|_{\mathcal{L}(X,Y)})$ *is Banach.*

*Proof.* Let $\{T_n\}_{n=1}^{\infty} \subseteq \mathcal{L}(X, Y)$ be Cauchy. Note that for any $x \in X$, $\{T_n(x)\}_{n=1}^{\infty}$ will be Cauchy in $Y$. Since $Y$ is complete, this sequence has a limit. Define $Tx := \lim_{n\to\infty} T_n x$. We must show that $T$ is linear, bounded, and $T_n \to T$.

Let $x, y \in X$, $\alpha \in \mathbb{F}$, and $n \in \mathbb{N}$. Then $T_n(x + \alpha y) = T_n x + \alpha T_n y$. By limit laws, $T(x + \alpha y) = Tx + \alpha Ty$. Thus $T$ is linear.

Since $\{T_n\}_{n=1}^{\infty}$ is Cauchy in $\mathcal{L}(X, Y)$, this means $\{T_n\}_{n=1}^{\infty}$ is bounded. Hence we can find $M > 0$ sufficiently large such that $\|T_n\| \leqslant M$ for all $n \in \mathbb{N}$. For every $x \in X$ with $\|x\| \leqslant 1$, we have:

$$\begin{aligned} \|Tx\| &= \left\| \lim_{n\to\infty} T_n x \right\| \\ &= \lim_{n\to\infty} \|T_n x\| \\ &\leqslant \lim_{n\to\infty} \|T_n\|_{\mathcal{L}(X,Y)} \|x\| \\ &\leqslant \lim_{n\to\infty} \|T_n\|_{\mathcal{L}(X,Y)} \\ &\leqslant M. \end{aligned}$$

Thus $T$ is bounded.

Let $\epsilon > 0$. Choose $N \in \mathbb{N}$ large such that $n, m \geqslant N$ implies $\|T_n - T_m\|_{\mathcal{L}(X,Y)} < \epsilon$. Let $x \in X$ such that $\|x\| \leqslant 1$. Then:

$$\begin{aligned} \|T_n x - T_m x\| &\leqslant \|(T_n - T_m)x\| \\ &\leqslant \|T_n - T_m\|_{\mathcal{L}(X,Y)} \|x\| \\ &\leqslant \|T_n - T_m\|_{\mathcal{L}(X,Y)} \\ &< \epsilon. \end{aligned}$$

Take $m \to \infty$. Then $\|T_n x - Tx\| \leqslant \epsilon$. Hence $\|(T_n - T)x\| \leqslant \epsilon$. Taking the supremum over all $x \in \overline{B}_X(0,1)$ gives $\|T_n - T\|_{\mathcal{L}(X,Y)} \leqslant \epsilon$. Thus $T_n \to T$ in the operator norm. □

**Corollary 1.4.1** (Neumann's Theorem). *Let $X$ be a Banach space and $T \in \mathcal{L}(X)$. If $\|T\| < 1$, then $I - T$ is invertible, and $(I-T)^{-1} = \sum_{n=0}^{\infty} T^n$.*

*Proof.* □

*Question.* For any normed linear space $X$, is $X^*$ nonempty? How can one construct an arbitrary bounded linear functional on $X$?

**Lemma 1.4.1** (Zorn's Lemma). *In a partially ordered set, if every chain has a maximal element, then the partially ordered set has a maximal element.*

**Theorem 1.4.2** (Hahn-Banach). *Let $X$ be a $\mathbb{R}$-vector space and $p : X \to \mathbb{R}$ a map satisfying:*

$$\begin{aligned} p(x+y) &\leqslant p(x) + p(y), \\ p(\lambda x) &= \lambda p(x), \text{ provided } \lambda > 0. \end{aligned}$$

*Let $Y \subseteq X$ be a subspace and $\varphi : Y \to \mathbb{R}$ a linear functional with:*

$$\varphi(y) \leqslant p(y) \text{ for all } y \in Y.$$

*Then there exists a linear functional $\Phi : X \to \mathbb{R}$ such that $\Phi|_Y = \varphi$ and:*

$$\Phi(x) \leqslant p(x) \text{ for all } x \in X.$$

*Proof.* □

**Corollary 1.4.2.** *Let $X$ be a normed linear space and $Y \subseteq X$ a subspace. If $\varphi \in Y^*$, then there exists an extension $\Phi \in X^*$ such that $\|\varphi\|_{Y^*} = \|\Phi\|_{X^*}$.*

*Proof.* Define $p : X \to \mathbb{R}$ by $p(x) := \|\varphi\|_{Y^*}\|x\|$. Clearly $p(x+y) \leqslant p(x) + p(y)$ and $p(\lambda x) = \lambda p(x)$ provided $\lambda > 0$. Moreover:

$$\begin{aligned} p(y) &= \|\varphi\|_{Y^*}\|y\| \\ &\geqslant |\varphi(y)| \\ &\geqslant \varphi(y). \end{aligned}$$

By the Hahn-Banach Theorem, there exists an extension $\Phi : X \to \mathbb{R}$ such that $\Phi|_Y = \varphi$ and $\Phi(x) \leqslant p(x)$. From this, we have the following inclusion of sets:

$$\left\{ \left|\varphi\left(\tfrac{y}{\|y\|}\right)\right| \mid y \in Y, y \neq 0 \right\} \subseteq \left\{ \left|\Phi\left(\tfrac{x}{\|x\|}\right)\right| \mid x \in X, x \neq 0 \right\}.$$

Hence $\|\varphi\|_{Y^*} \leqslant \|\Phi\|_{X^*}$. Conversely, by our construction of $p$ we have:

$$\begin{aligned} |\Phi(x)| &\leqslant |p(x)| \\ &\leqslant \|\varphi\|_{Y^*} \|x\| . \end{aligned}$$

Rearranging gives $\left|\Phi\left(\frac{x}{\|x\|}\right)\right| \leqslant \|\varphi\|_{Y^*}$. Taking the supremum over all $x \in X$, $x \neq 0$ gives $\|\Phi\|_{X^*} \leqslant \|\varphi\|_{Y^*}$. Having established equality, we've finally shown $\Phi \in X^*$. □

**Corollary 1.4.3.** *Let $X$ be a normed linear space and $Y \subseteq X$ a subspace. If $w \in X \setminus Y$ such that $\operatorname{dist}(w, Y) > 0$, then there exists a bounded linear functional $\varphi \in X^*$ such that $\varphi(w) = 1$, $\varphi(y) = 0$ for all $y \in Y$, and $\|\varphi\|_{X^*} = \frac{1}{\operatorname{dist}(w,Y)}$.*

*Proof.* Define $\varphi : \operatorname{span}\{Y, \{w\}\} \to \mathbb{R}$ by $\varphi(y + \lambda w) = \lambda$. This is a linear functional, and:

$$\begin{aligned} \|\varphi\| &= \sup\left\{\left|\varphi\left(\tfrac{y+\lambda w}{\|y+\lambda w\|}\right)\right| : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0\right\} \\ &= \sup\left\{\frac{|\lambda|}{\|y + \lambda w\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0\right\} \\ &= \sup\left\{\frac{1}{\left\|\frac{y}{\lambda} + w\right\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0\right\} \\ &= \sup\left\{\frac{1}{\|w - y\|} : y \in Y, \lambda \in \mathbb{R}, y + \lambda w \neq 0\right\}^1 \\ &= \frac{1}{\inf\{\|w - y\| \mid y \in Y\}}^2 \\ &= \frac{1}{\operatorname{dist}(w, Y)}. \end{aligned}$$

Applying Corollary 1.4.2 establishes the proof. □

*Remark.* The previous corollary establishes that if $X = \{0\}$, then $X^* \neq \{0\}$.

## Exercises

**Exercise 1.4.1.** *Let $X$ be a normed linear space. Prove the limit laws: if $x_n \to x$, $y_n \to y$, and $\lambda \in \mathbb{F}$, then $x_n + \lambda y_n \to x + \lambda y$.*

*Proof.* □

**Exercise 1.4.2.** *Let $X$ be a normed linear space with norm $\|\cdot\|$. Prove that the map $\|\cdot\| : X \to \mathbb{R}$ is continuous.*

[1] Since $\{\frac{y}{\lambda} \mid y \in Y, \lambda \in \mathbb{R}\} = Y$.

[2] Let $x_n \to \inf A$. Then $\frac{1}{x_n} \to \frac{1}{\inf A}$. Since $\frac{1}{\inf A}$ is an upper-bound and an adherent point of $\frac{1}{A}$, we have $\frac{1}{\inf A} = \sup \frac{1}{A}$.

*Proof.* Let $x, y \in X$. Then $\|x\| \leqslant \|x-y\| + \|y\|$, giving $\|x\| - \|y\| \leqslant \|x-y\|$. A similar tactic gives $\|y\| - \|x\| \leqslant \|x-y\|$. Hence $|\|x\| - \|y\|| \leqslant \|x-y\|$. This means $\|\cdot\|$ is Lipschitz, implying that it is continuous. □

**Exercise 1.4.3.** *Prove that the closure of a linear subspace in a normed linear space is also a subspace.*

*Proof.* Let $X$ be a normed linear space and $Y$ a subspace. Let $u, v \in \overline{Y}$ and $\alpha \in \mathbb{F}$. There exists sequences $\{u_n\}_{n=1}^{\infty}, \{v_n\}_{n=1}^{\infty}$ such that $u_n \to u$ and $v_n \to v$. Hence $u_n + \alpha v_n \to u + \alpha v \in \overline{Y}$. Thus $\overline{Y}$ is a linear subspace. □

**Exercise 1.4.4.** *Prove that the operator norm has the three properties required of a norm*

*Proof.* □

**Exercise 1.4.5.** *Prove directly that if $T$ is an unbounded linear operator, then it is discontinuous at 0.*

*Proof.* Let $T : X \to Y$ be an unbounded linear operator. Since $T$ is unbounded:

$$\|T\|_{\mathcal{L}(X,Y)} = \sup\{\|Tx\|_Y \mid x \in \overline{B}_X(0,1)\} = \infty.$$

We can find a sequence $\{x_n\}_{n=1}^{\infty}$ such that $\|x\|_X \leqslant 1$ and $\|Tx_n\|_Y \to \infty$. Consider the sequence $\left\{\frac{x_n}{\|Tx_n\|_Y}\right\}_{n=1}^{\infty}$. Clearly $\frac{x_n}{\|Tx_n\|_Y} \to 0$. But:

$$\left\|T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\| = \frac{\|Tx_n\|_Y}{\|Tx_n\|_Y} = 1.$$

Hence the sequence $\left\{T\left(\frac{x_n}{\|Tx_n\|_Y}\right)\right\}_{n=1}^{\infty}$ does *not* converge to 0. Thus $T$ is discontinuous at 0. □

**Exercise 1.4.6.** *Prove that the norm of a linear transformation is the infimum of all numbers $M$ that satisfy the inequality $\|Tx\| \leqslant M\|x\|$ for all $x$.*

*Proof.* Let $M$ be such that $\|Tx\| \leqslant M\|x\|$ for all $x \in X$. Then $\frac{\|Tx\|}{\|x\|} \leqslant M$ for all $x \in X$, $x \neq 0$. Taking the supremum over all $x \in X, x \neq 0$ gives $\|T\|_{\mathcal{L}(X,Y)} \leqslant M$. Hence $\|T\|_{\mathcal{L}(X,Y)} \leqslant \inf\{M \mid \|Tx\| \leqslant M\|x\| \text{ for all } x \in X\}$.

Conversely, let $x \in X, x \neq 0$. Then $\|Tx\| \leqslant \|T\|_{\mathcal{L}(X,Y)}\|x\|$. Thus $\|T\|_{\mathcal{L}(X,Y)} \in \{M \mid \|Tx\| \leqslant M\|x\| \text{ for all } x \in X\}$. Surely $\|T\|_{\mathcal{L}(X,Y)} \geqslant \inf\{M \mid \|Tx\| \leqslant M\|x\| \text{ for all } x \in X\}$, establishing equality. □

**Exercise 1.4.7.** *On the space $C([0,1])$ we define "point-evaluation functionals" by $t^*(f) = f(t)$. Here $t \in [0,1]$ and $f \in C([0,1])$. Prove that $\|t^*\| = 1$. Prove that if $\varphi = \sum_{i=1}^{n} \lambda_i t_i^*$, where $t_1, t_2, ..., t_n$ are distinct points in $[0,1]$, then $\|\varphi\| = \sum_{i=1}^{n} |\lambda_i|$.*

*Proof.* □

**Exercise 1.4.8.** *Prove that if a linear transformation maps some nonvoid open set of the domain space to a bounded set in the range space, then it is continuous.*

*Proof.* Let $T : X \to Y$ be linear. Let $U \subseteq X$ be open and nonvoid. Then there exists some $x \in U$. Find $r > 0$ such that $B(x, r) \subseteq U$.

Let $x_0 \in B(0, 1)$. Then $x + rx_0 \in B(x, r) \subseteq U$. Hence $T(x + rx_0) \in T(U)$. We can write:

$$\|Tx_0\| \leqslant \frac{\|T(x + rx_0)\| + \|Tx\|}{r}.$$

Since $T(U)$ is bounded, it must be the case that $\|Tx_0\| \leqslant M$ for some $M > 0$. Thus $T(B(0, 1))$ is bounded.

Now let $x \in X$. We have that:

$$\begin{aligned} \|Tx\| &= \left\| T\left(\frac{x}{2\|x\|} \cdot 2\|x\|\right) \right\| \\ &= 2\left\| T\left(\frac{x}{2\|x\|}\right) \right\| \|x\| \\ &\leqslant 2M\|x\|. \end{aligned}$$

Thus $T$ is continuous. □

**Exercise 1.4.9.** *Let $Y$ be a linear subspace in a normed linear space $X$. For a fixed $x \in X$, prove that:*

$$\operatorname{dist}(x, Y) = \sup\{\varphi(x) \mid \varphi \in X^*, \ \varphi\big|_Y \equiv 0, \ \|\varphi\| = 1\}.$$

*Proof.* Let $\varphi \in X^*$ such that $\varphi|_Y = 0$ and $\|\varphi\| = 1$. Observe that:

$$\begin{aligned} \varphi(x) &= \varphi(x) - \varphi(y) \\ &= \varphi(x - y) \\ &\leqslant \|\varphi\| \|x - y\| \\ &= \|x - y\|. \end{aligned}$$

Taking the infimum over all $y \in Y$ gives $\varphi(x) \leqslant \operatorname{dist}(x, Y)$. Taking the supremum over all $\varphi \in X^*$ such that $\varphi|_Y = 0$ and $\|\varphi\| = 1$ gives $\sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\} \leqslant \operatorname{dist}(x, Y)$.

For the converse direction, we proceed by cases on $\operatorname{dist}(x, Y)$. If $\operatorname{dist}(x, Y) = 0$, then by the above inequality we have:

$$\begin{aligned} 0 &= \operatorname{dist}(x, Y) \\ &\geqslant \sup\{\varphi(x) \mid \varphi \in X^*, \varphi\big|_Y = 0, \|\varphi\| = 1\} \\ &\geqslant 0. \end{aligned}$$

Hence $\operatorname{dist}(x, Y) = \sup\{\varphi(x) \mid \varphi \in X^*, \varphi|_Y = 0, \|\varphi\| = 1\}$. Now suppose $\operatorname{dist}(x, Y) > 0$. By Corollary 1.4.3 there exists a linear functional $\psi \in X^*$ such that $\psi(x) = 1$, $\psi|_Y = 0$, and

$\|\psi\| = \frac{1}{\text{dist}(x,Y)}$. Define a linear functional $\Phi : X \to \mathbb{F}$ by $\Phi(x_0) = \text{dist}(x_0, Y)\psi(x_0)$. Note that $\Phi|_Y = 0$ and:

$$\|\|$$

$\|\Phi\| = \text{dist}()$ □

## § 1.5. $\ell^p$ Spaces

**Definition 1.5.1.**

(1) Let $1 \leqslant p < \infty$. We define $\ell^p := \{x : \mathbb{N} \to \mathbb{F} \mid \sum_{j=1}^{\infty} |x_j|^p < \infty\}$. Likewise, define $\|\cdot\|_{\ell^p} : \ell^p \to \mathbb{F}$ by $\|x\|_{\ell^p} = \left(\sum_{j=1}^{\infty} |x_j|^p\right)^{\frac{1}{p}}$.

(2) Let $p = \infty$. We define $\ell^\infty := \{x : \mathbb{N} \to \mathbb{F} \mid \sup_{n\in\mathbb{N}} |x_n| < \infty\}$. Likewise, define $\|\cdot\|_{\ell^\infty} : \ell^\infty \to \mathbb{F}$ by $\|x\|_{\ell^\infty} = \sup_{n\in\mathbb{N}} |x_n|$.

**Lemma 1.5.1.** *Let* $a, b \geqslant 0$ *and* $0 \leqslant t \leqslant 1$. *Then* $((1-t)a + tb)^p \leqslant (1-t)a^p + tb^p$.

*Proof.* □

**Lemma 1.5.2.** *Let* $a, b \geqslant 0$ *and* $0 \leqslant t \leqslant 1$. *Then* $(a+b)^p \leqslant 2^{p-1}(a^p + b^p)$.

*Proof.* □

**Lemma 1.5.3** (Minkowski's Inequality)**.** *Let* $1 \leqslant p \leqslant \infty$. *For any* $x, y \in \ell^p$, *we have* $\|x + y\|_{\ell^p} \leqslant \|x\|_{\ell^p} + \|y\|_{\ell^p}$.

**Theorem 1.5.1.** *For any* $1 \leqslant p < \infty$, $(\ell^p, \|\cdot\|_{\ell^p})$ *is a Banach space.*

*Proof.* Lemma 1.5.1 establishes that $\ell^p$ is a vector space. Lemma 1.5.3 established that $\|\cdot\|_{\ell^p}$ satisfies the triangle inequality; the remaining norm axioms are routine to check. Hence $(\ell^p, \|\cdot\|_{\ell^p})$ is a normed linear space.

Let $\{f_n\}_{n=1}^{\infty} \subseteq \ell^p$ be a Cauchy sequence. For any $j, n, m \in \mathbb{N}$, we have:

$$\begin{aligned} |f_n(j) - f_m(j)| &\leqslant (|f_n(j) - f_m(j)|^p)^{\frac{1}{p}} \\ &\leqslant \left(\sum_{k=1}^{\infty} |f_n(k) - f_m(k)|^p\right)^{\frac{1}{p}} \\ &= \|f_n - f_m\|_{\ell^p}. \end{aligned}$$

Hence $\{f_n(j)\}_{n=1}^{\infty} \subseteq \mathbb{F}$ is Cauchy. Since $\mathbb{F}$ is complete, the limit of this sequence exists. Define $f(j) := \lim_{n\to\infty} f_n(j)$. We will show that $f \in \ell^p$ and that $f_n \to f$ in $\ell^p$.

Find $N \in \mathbb{N}$ sufficiently large such that for all $n \geqslant N$, we have $\|f_n - f_N\|_{\ell^p} \leqslant 1$. Taking $p^{\text{th}}$ powers on both sides gives the following statement: for all $n \geqslant N$ we have $\sum_{j=1}^{\infty} |f_n(j) - f_N(j)|^p \leqslant 1$. Now fix $J \in \mathbb{N}$. For all $n \geqslant N$ we have $\sum_{j=1}^{J} |f_n(j) - f_N(j)|^p \leqslant 1$. Take $n \to \infty$. Then we have $\sum_{j=1}^{J} |f(j) - f_N(j)|^p \leqslant 1$. In particular, for all $J \in \mathbb{N}$ we have:

$$\begin{aligned}
\sum_{j=1}^{J} |f(j)|^p &= \sum_{j=1}^{J} |f(j) - f_N(j) + f_N(j)|^p \\
&\leqslant \sum_{j=1}^{J} 2^{p-1} \left(|f(j) - f_N(j)|^p + |f_N(j)|^p\right) \qquad \text{Lemma 1.5.2} \\
&= 2^{p-1} \left( \sum_{j=1}^{J} |f(j) - f_N(j)|^p + \sum_{j=1}^{J} |f_N(j)|^p \right) \\
&\leqslant 2^{p-1}(1 + \|f_N\|_{\ell^p}^p) \\
&< \infty.
\end{aligned}$$

By the Monotone Convergence Theorem, $\sum_{j=1}^{\infty} |f(j)|^p < \infty$. Hence $f \in \ell^p$.

Now let $\epsilon > 0$. Choose $N \in \mathbb{N}$ such that $\|f_n - f_N\|_{\ell^p} \leqslant \epsilon$. Then for all $n \geqslant N$ and for all $j \in \mathbb{N}$, we have $\sum_{j=1}^{J} |f_n(j) - f_N(j)|^p \leqslant \epsilon^p$. Take $n \to \infty$. Then for all $J \in \mathbb{N}$ we have $\sum_{j=1}^{J} |f(j) - f_N(j)|^p \leqslant \epsilon^p$. Using the Monotone Convergence Theorem, take $J \to \infty$. Then $\sum_{j=1}^{\infty} |f(j) - f_N()$ Then $\sum_{j=1}^{\infty} |f(j) - f_N(j)|^p \leqslant \epsilon^p$. Therefore $\|f - f_N\|_{\ell^p} \leqslant \epsilon$. □

**Definition 1.5.2.** The subspace of $\ell^\infty$ consisting of all convergent sequences is denoted $c := \{x \in \ell^\infty \mid x \text{ converges}\}$. The subspace of $c$ consisting of all sequences which converge to zero is denoted $c_0 := \{x \in c \mid x \text{ converges to } 0\}$. The subspace of $\ell^\infty$ consisting of all sequences with only finitely many nonzero terms is denoted $c_{00} := \{x \in \ell^\infty \mid \text{supp}(x) < \infty\}$.

**Definition 1.5.3.** The standard coordinate vector in an infinite dimensional vector space is denoted $e^n := (0, ..., 0, 1, 0, ...)$; i.e., it is a sequence which is "1" in the $n^{\text{th}}$ coordinate and "0" elsewhere.

**Proposition 1.5.1.** *Let $y \in \ell^\infty$. Define $\varphi_y : \ell^1 \to \mathbb{F}$ by $\varphi_y(x) = \sum_{k=1}^{\infty} x_k y_k$. Then $\varphi \in (\ell^1)^*$ and $\|\varphi_y\|_{\ell^1}^* = \|y\|_{\ell^\infty}$.*

## Exercises

**Exercise 1.5.1.** *Show that $(\ell^\infty, \|\cdot\|_\infty)$ is complete.*

*Proof.* □

**Exercise 1.5.2.** *Show that $c$ and $c_0$ are closed in $\ell^\infty$. Show that $c_{00}$ is not closed by exhibiting a sequence in $c_{00}$ which converges to a point $x \in \ell^\infty$ but is not in $c_{00}$.*

*Proof.* □

**Exercise 1.5.3.** *Let $y \in \ell^\infty$ and define the linear functional:*

$$\varphi_y : \ell^1 \to \mathbb{F}, \quad \varphi_y(x) := \sum_{j=1}^{\infty} x_j y_j.$$

*Show that* $\|\varphi_y\|_{(\ell^1)^*} = \|y\|_{\ell^\infty}$.

*Proof.* Note that:

$$\begin{aligned} \sum_{i=1}^{\infty} |x_i y_i| &= \sum_{i=1}^{\infty} |x_i||y_i| \\ &\leqslant \|y\|_{\ell^\infty} \sum_{i=1}^{\infty} |x_i| \\ &\leqslant \|y\|_{\ell^\infty} \|x\|_{\ell^1}. \end{aligned}$$

Hence:

$$\begin{aligned} |\varphi_y(x)| &= \left| \sum_{i=1}^{\infty} x_i y_i \right| \\ &\leqslant \sum_{i=1}^{\infty} |x_i y_i| \\ &\leqslant \|y\|_{\ell^\infty} \|x\|_{\ell^1}. \end{aligned}$$

Dividing by $\|x\|_{\ell^1}$ and taking the supremum over all $x \in \ell^1$, $x \neq 0$ gives the inequality $\|\varphi_y\|_{(\ell^1)^*} \leqslant \|y\|_{\ell^\infty}$. Hence $\varphi_y \in (\ell^1)^*$.

For the converse direction, note that $e_j \in B_{\ell^1}(0,1)$. Hence for any $j \in \mathbb{N}$, we have $|y_j| \leqslant |\varphi_y(e_j)| \leqslant \|\varphi_y\|_{(\ell^1)^*}$. Taking the supremum over all $j \in J$ gives $\|y\|_{\ell^\infty} \leqslant \|\varphi_y\|_{(\ell^1)^*}$. Thus $\|\varphi_y\|_{(\ell^1)^*} = \|y\|_{\ell^\infty}$. □

## § 1.6. Consequences of The Baire Category Theorem

**Theorem 1.6.1** (Baire Category Theorem). *Let $X$ be a complete metric space. If $\{U_n\}_{n=1}^{\infty}$ is a family of open and dense subsets of $X$, then $\bigcap_{n=1}^{\infty} U_n$ is still dense.*

*Proof.* □

**Corollary 1.6.1.** *Let $X$ be a complete metric space and $\{Z_n\}_{n=1}^{\infty}$ a family of closed sets. If $X = \bigcup_{n=1}^{\infty} Z_n$, then there exists some $i \in \mathbb{N}$ such that $Z_i$ has non-empty interior.*

*Proof.* □

**Definition 1.6.1.** Let $X$ be a metric space.

(1) A subset $Y$ of $X$ is called <u>*nowhere dense*</u> if $\overline{Y}$ has empty interior.

(2) A subset of $X$ is called <u>*Category I*</u> if it is the countable union of nowhere dense sets.

(3) A subset of $X$ which is not Category I is called <u>*Category II*</u>.

*Remark.* Theorem 1.6.1 and Corollary 1.6.1 can be rephrased as the assertion that every complete metric space is Category II.

**Theorem 1.6.2** (Banach-Steinhouse)**.** *Let $X$ be a Banach space and $Y$ a normed linear space. Let $\{T_\alpha\}_{\alpha\in A} \subset \mathcal{L}(X, Y)$ be an arbitrary family of bounded linear operators. Then $\sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$ if and only if $\{x \mid \sup_{\alpha\in A} \|T_\alpha x\| < \infty\}$ is Category II*

*Proof.* Suppose $C := \sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$. Then for all $x \in X$ and $\alpha \in A$, we have $\|T_\alpha x\|_Y \leqslant \|T_\alpha\| \|x\|_X \leqslant C\|x\|$. Hence $\{x \mid \sup_{\alpha\in A} \|T_\alpha x\| < \infty\} = X$, which is Category II by the Baire Category Theorem.

Conversely, suppose $F := \{x \mid \sup_{\alpha\in A} \|T_\alpha x\| < \infty\}$ is Category II. Define $F_n := \{x \mid \sup_{\alpha\in A} \|T_\alpha x\|\}$. We will show that $F_n$ is closed in $X$. Note that:

$$F_n = \bigcap_{\alpha\in A} \{x \mid \|T_\alpha x\| \leqslant n\}.$$

Note that $\|\cdot\|$ and $T_\alpha$ are continuous, hence the composition $g_\alpha := \|T_\alpha(\cdot)\|$ is continuous. The above expression then becomes:

$$\begin{aligned} F_n &= \bigcap_{\alpha\in A} \{x \mid \|T_\alpha x\| \leqslant n\} \\ &= \bigcap_{\alpha\in A} \{x \mid g_\alpha(x) \leqslant n\} \\ &= \bigcap_{\alpha\in A} \{x \mid g_\alpha(x) \in [0, n]\} \\ &= \bigcap_{\alpha\in A} g_\alpha^{-1}([0, n]). \end{aligned}$$

Hence $F_n$ is closed.

Since $F = \bigcup_{n=1}^{\infty} F_n$, by Corollary 1.6.1 there exists some $n \in \mathbb{N}$ such that $F_n$ has nonempty interior. Choose $x_0 \in X$ and $r > 0$ such that $\overline{B}(x_0, r) \subset F_n$. Let $x \in X$ such that $\|x\| \leqslant 1$. Fix $\alpha \in A$. Note that $x_0 + rx \in \overline{B}(x_0, r)$, hence:

$$\begin{aligned} \|T_\alpha x\|_Y &= \left\| T_\alpha \left( \frac{1}{r} (x_0 + rx - x_0) \right) \right\|_Y \\ &\leqslant \frac{1}{r} (\|T_\alpha(x_0 + rx)\|_Y + \|T_\alpha x_0\|_Y) \\ &\leqslant \frac{2n}{r} \\ &< \infty. \end{aligned}$$

Taking the supremum over all $\alpha$ and $\|x\| \leqslant 1$ gives $\sup\{\|T_\alpha\| \mid \alpha \in A\} < \infty$. □

**Corollary 1.6.2** (Uniform Boundedness Principle)**.** *Let* $X$ *be a Banach space and* $Y$ *a normed linear space. Let* $\{T_\alpha\}_{\alpha \in A} \subset \mathcal{L}(X, Y)$ *be an arbitrary family of bounded linear operators. If* $\sup_{\alpha\in A} \|T_\alpha x\|_Y < \infty$ *for all* $x \in X$*, then* $\sup_{\alpha\in A} \|T_\alpha\| < \infty$*.*

*Proof.* The set $\{x \mid \sup_{\alpha\in A} \|T_\alpha x\| < \infty\} = X$ is Category II. By Theorem 1.6.2 we have that $\sup_{\alpha\in A} \|T_\alpha\| < \infty$. □

**Definition 1.6.2.** Let $X$ and $Y$ be normed linear spaces and $T : X \to Y$ linear.

(1) The graph of $T$ is $\mathrm{Gr}(T) := \{(x, Tx) \mid x \in X\}$.

(2) The map $T$ has a closed graph if $\mathrm{Gr}(T)$ is closed in $X \times Y$.

*Remark.* Regardless if $X \times Y$ is given the product or box topology, the induced metrics are equivalent.

**Proposition 1.6.1.** *Let* $X, Y$ *be normed spaces and* $T : X \to Y$ *be linear. The following are equivalent:*

*(1)* $T$ *has a closed graph;*

*(2) If* $\{x_n\}_{n=1}^\infty$ *is a sequence such that* $x_n \to x$ *and* $Tx_n \to y$*, then* $y = Tx$*.*

*Proof.* Suppose $T$ has a closed graph. Let $\{x_n\}_{n=1}^\infty$ be a sequence such that $x_n \to x$ and $Tx_n \to y$. Then $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \mathrm{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \to (x, y)$. But since $T$ has a closed graph, $(x, y) \in \mathrm{Gr}(T)$. Hence $y = Tx$.

Conversely, if $\{(x_n, Tx_n)\}_{n=1}^\infty \subseteq \mathrm{Gr}(T)$ is a sequence such that $(x_n, Tx_n) \to (x, y) \in X \times Y$, then $x_n \to x$ and $Tx_n \to y$. By hypothesis $y = Tx$, hence $(x, y) \in \mathrm{Gr}(T)$. Thus $T$ has a closed graph. □

*Remark.* A linear map $T$ having a closed graph is weaker than $T$ being continuous.

**Example 1.6.1.** Recall the following standard result from an introductory analysis course: let $J \subseteq \mathbb{R}$ be a bounded interval and let $\{f_n\}_{n=1}^\infty \subseteq J^{\mathbb{R}}$ be a sequence of functions. Suppose that there exists $x_0 \in J$ such that $\{f_n(x_0)\}_{n=1}^\infty$ converges, and that the sequence $\{f'_n\}_{n=1}^\infty$ of derivatives exists on $J$ and converges uniformly on $J$ to a function $g$. Then the sequence $\{f_n\}_{n=1}^\infty$ converges uniformly on $J$ to a function $f$ that has a derivative at every point of $J$ and $f' = g$.

Indeed, define $D : C^1([0, 1]) \to C([0, 1])$ by $Df = f'$. Let $\{f_n\}_{n=1}^\infty \subseteq C^1([0, 1])$ be a sequence of functions such that $f_n \to f$ and $Df_n \to g$. Note that the convergence of $\{Df_n\}_{n=1}^\infty$ will be uniform by the compactness of $[0, 1]$. Hence by the above result, we have that $g = Df$. By Proposition 1.6.1, $D$ has a closed graph. However, we showed in Example 1.3.3 that $D$ is not continuous.

**Definition 1.6.3.** Let $X, Y$ be normed linear spaces. A map $T : X \to Y$ is open if $T(U)$ is open in $Y$ provided $U$ is open in $X$.

**Theorem 1.6.3** (Open-Mapping Theorem)**.** *Let* $T : X \to Y$ *be a continuous, surjective, and linear mapping between Banach spaces. Then* $T$ *is open.*

*Remark.* Many times throughout the following proof we will be using the fact that $B(0,n) = nB(0,1)$.

*Proof.* Note that $Y = \bigcup_{n=1}^{\infty} T(nB_X(0,1))$, hence $Y = \bigcup_{n=1}^{\infty} \overline{T(nB_X(0,1))}$. Since $Y$ is Banach, the Baire Category Theorem says there exists some $n \in \mathbb{N}$ such that $\overline{nT(B_X(0,1))}$ has nonempty interior. So there exists $y_0 \in Y$ and $r > 0$ such that:

$$\begin{aligned} B_Y(y_0, r) &\subset \overline{nT(B_X(0,1))} \\ &= n\overline{T(B_X(0,1))}^3. \end{aligned}$$

Let $\epsilon > 0$. Since $y_0 \in B_Y(y_0, r)$, surely $y_0 \in n\overline{T(B_X(0,1))}$. Hence there exists some $x_1 \in B_X(0,1)$ such that $\|y_0 - nTx_1\|_Y < \epsilon$. Now let $y \in B(0,1)$ be arbitrary. Then $y_0 + ry \in B_Y(y_0, r) \subset n\overline{T(B_X(0,1))}$. Hence there exists $x_2 \in B_X(0,1)$ such that $\|y_0 + ry - nTx_2\|_Y < \epsilon$. We can write:

$$\begin{aligned} y &= \frac{1}{r}(ry + y_0) - \frac{1}{r}y_0 \\ &= \frac{1}{r}(ry + y_0 - nTx_2) + \frac{n}{r}Tx_2 - \frac{1}{r}(y_0 - nTx_1) - \frac{n}{r}Tx_1 \\ &= \frac{1}{r}\left((y_0 + ry) - nTx_2\right) - \frac{1}{r}(y_0 - nTx_1) + \frac{n}{r}T(x_2 - x_1). \end{aligned}$$

We can rearrange the above equation as:

$$y - \frac{n}{r}T(x_2 - x_1) = \frac{1}{r}\left((y_0 + ry) - nTx_2\right) - \frac{1}{r}(y_0 - nTx_1).$$

Hence:

$$\begin{aligned} \left\|y - \frac{n}{r}T(x_2 - x_1)\right\| &= \left\|\frac{1}{r}\left((y_0 + ry) - nTx_2\right) - \frac{1}{r}(y_0 - nTx_1)\right\| \\ &\leqslant \frac{1}{r}\left\|((y_0 + ry) - nTx_2)\right\| + \frac{1}{r}\left\|(y_0 - nTx_1)\right\| \\ &< \frac{2\epsilon}{r}. \end{aligned}$$

Take $x' := x_2 - x_1 \in B(0,2)$. Then since $\left\|y - \frac{n}{r}T(x')\right\| < \frac{2\epsilon}{r}$ can be made arbitrarily small, we have that $y \in \overline{T(B_X(0, \frac{2n}{r}))}$. Take $\lambda := \frac{2n}{r}$. Thus we've established the inclusion $B_Y(0,1) \subset \overline{TB_X(0,\lambda)}$.

[3] Follows from the fact that $\overline{B(x,r)} = \overline{B}(x,r)$ and the above remark.

Let $y \in B_Y(0,1) \subset \overline{T(B_X(0,\lambda))}$. Find $x_1 \in B_X(0,\lambda)$ such that $\|y - Tx_1\|_Y < \frac{1}{2}$. Then:

$$\begin{aligned} y - Tx_1 &\in B_Y\left(0, \frac{1}{2}\right) \\ &= \frac{1}{2} B_Y(0,1) \\ &\subset \frac{1}{2}\overline{T(B_X(0,\lambda))} \\ &= \overline{\frac{1}{2}T(B_X(0,\lambda))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2}\right)\right)}. \end{aligned}$$

Now pick $x_2 \in B_X(0, \frac{\lambda}{2})$ such that $\|y - Tx_1 - Tx_2\| < \frac{1}{2^2}$. Then:

$$\begin{aligned} y - Tx_1 - Tx_2 &\in B_Y(0, \frac{1}{2^2}) \\ &= \frac{1}{2^2} B_Y(0,1) \\ &\subset \frac{1}{2^2}\overline{T(B_X(0,\lambda))} \\ &= \overline{\frac{1}{2}T(B_X(0,1))} \\ &= \overline{T\left(B_X\left(0, \frac{\lambda}{2^2}\right)\right)}. \end{aligned}$$

Inductively, pick $x_k \in B_X(0, \frac{\lambda}{2^{k-1}})$ such that $\left\|y - \sum_{j=1}^k Tx_j\right\| < \frac{1}{2^k}$. Note that for all $k \in \mathbb{N}$, we have $\|x_k\|_X < \frac{\lambda}{2^{k-1}}$. Therefore $\sum_{j=1}^\infty x_j$ is absolutely convergent. Since $X$ is Banach, by Proposition 1.1.2 the series $\sum_{j=1}^\infty x_j$ is convergent. In particular:

$$\begin{aligned} \left\|\sum_{j=1}^\infty x_j\right\|_X &\leqslant \sum_{j=1}^\infty \|x_j\|_X \\ &< \sum_{j=1}^\infty \frac{\lambda}{2^{j-1}} \\ &= 2\lambda. \end{aligned}$$

Hence $\sum_{j=1}^\infty x_j \in B_X(0,2\lambda)$. Now note that $\sum_{j=1}^k x_j \to \sum_{j=1}^\infty x_j$ in $X$. Also, $\left\|y - \sum_{j=1}^k Tx_j\right\| < \frac{1}{2^k}$ implies that $T\left(\sum_{j=1}^k x_k\right) \to y$ in $Y$. Since $T$ is continuous, $T$ has a closed graph. Hence $y = T\sum_{j=1}^\infty x_j \in T(B_X(0,2\lambda))$. Since $y \in B_Y(0,1)$ was arbitrary, we've established the inclusion $B_Y(0,1) \subset T(B_X(0,2\lambda))$.

Now let $U \subseteq X$ be open. Let $y_0 \in T(U)$. Since $T$ is surjective, find $x_0 \in U$ such that

$Tx_0 = y_0$. Since $U$ is open there exists an $\epsilon > 0$ such that $B_X(x_0, \epsilon) \subseteq U$. Therefore:

$$\begin{aligned} T(U) &\supset T(B_X(x_0, \epsilon)) \\ &= Tx_0 + \frac{\epsilon}{2\lambda} T(B_X(0, 2\lambda)) \\ &\supset y_0 + \frac{\epsilon}{2\lambda} B_Y(0,1) \\ &= B_Y\left(y_0, \frac{\epsilon}{2\lambda}\right) \end{aligned}$$

Thus $T(U)$ is open, establishing $T$ as an open map. □

**Corollary 1.6.3.** *Let $T : X \to Y$ be a bijective, linear, and continuous map between Banach spaces. Then $T^{-1}$ is continuous.*

*Proof.* Since $T$ is continuous, $T$ has a closed graph. By the Open-Mapping Theorem $T$ is an open map. Hence $(T^{-1})^{-1}(U) = T(U)$ is open. □

**Corollary 1.6.4.** *Let $X$ be a vector space. Suppose $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms such that $(X, \|\cdot\|_\alpha)$ and $(X, \|\cdot\|_\beta)$ are Banach. If there exists a $C > 0$ such that $\|x\|_\alpha \leqslant C\|x\|_\beta$ for all $x \in X$, then there exists $C' > 0$ such that $\|x\|_\beta \leqslant C'\|x\|_\alpha$ for all $x \in X$.*

*Proof.* Certainly the map $\mathrm{id} : (X, \|\cdot\|_\beta) \to (X, \|\cdot\|_\alpha)$ is bijective and linear. Moreover, $\|\mathrm{id}\, x\|_\alpha = \|x\|_\alpha \leqslant C\|x\|_\beta$. Hence id is continuous. By Corollary 1.6.3 $\mathrm{id}^{-1}$ is continuous. Hence there exists $C' > 0$ such that $\|x\|_\beta = \left\|\mathrm{id}^{-1} x\right\|_\beta \leqslant C'\|x\|_\alpha$. □

*Remark.* The above corollary can be rephrased as: if $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are norms such that $(X, \|\cdot\|_\alpha)$ and $(X, \|\cdot\|_\beta)$ are Banach, then $\|\cdot\|_\alpha$ and $\|\cdot\|_\beta$ are equivalent.

**Theorem 1.6.4** (Closed Graph Theorem)**.** *Let $X$ and $Y$ be Banach spaces. If $T : X \to Y$ is linear and has a closed graph, then $T$ is continuous.*

*Proof.* Define $\|\cdot\|_* : X \to \mathbb{R}$ by $\|x\|_* := \|x\|_X + \|Tx\|_Y$. It is routine to check that $(X, \|\cdot\|_*)$ is a normed linear space. We will check that $(X, \|\cdot\|_*)$ is Banach. Suppose $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_*$. Then clearly $\{x_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_X$ and $\{Tx_k\}_{k=1}^\infty$ is Cauchy with respect to $\|\cdot\|_Y$. Since $X$ and $Y$ are Banach, there exists $x \in X$ and $y \in Y$ such that $x_k \to x$ with respect to $\|\cdot\|_X$ and $Tx_k \to y$ with respect to $\|\cdot\|_Y$. Since $T$ has a closed graph, $y = Tx$. Hence:

$$\begin{aligned} \|x - x_k\|_* &= \|x - x_k\|_X + \|Tx - Tx_k\|_Y \\ &\to 0 \text{ as } k \to \infty. \end{aligned}$$

Thus $x_k \to x$ with respect to $\|\cdot\|_*$, establishing that $(X, \|\cdot\|_*)$ is Banach. But since $(X, \|\cdot\|_X)$ and $(X, \|\cdot\|_*)$ are Banach, and since $\|x\|_X \leqslant \|x\|_*$, Corollary 1.6.4 says there exists $C' > 0$ such that $\|x\|_* \leqslant C'\|x\|_X$. Therefore $\|Tx\|_Y \leqslant \|x\|_* \leqslant C'\|x\|_X$ for all $x \in X$. Thus $T$ is continuous. □

**Example 1.6.2.** jan 29th, bijective but inverse is unbounded

**Example 1.6.3.** bounded but inverse is unbounded

## Exercises

**Exercise 1.6.1.** *Let $f$ be infinitely differentiable on $\mathbb{R}$. Suppose for any $x \in \mathbb{R}$ there is some $n$ such that $f^{(n)}(x) = 0$ (note that $n$ may depend on $x$). Show that $f$ must be a polynomial. Hint: Consider the sets $Z_n = \{x \mid f^{(n)}(x) = 0\}$ and recall Baire's Category Theorem.*

*Proof.* □

**Exercise 1.6.2.** *Let $T : X \longrightarrow Y$ be a linear map between normed spaces. Prove the assertion made in class: The graph $\{(x, Tx) \mid x \in X\}$ of $T$ is closed in $X \times Y$ if and only if:*

$$\text{if } x_n \longrightarrow x \text{ and } Tx_n \longrightarrow y \text{ then } y = Tx.$$

*Proof.* □

**Exercise 1.6.3.** *Define $T : c_0 \longrightarrow c_0$ by $T(\{x_n\}_{n=1}^{\infty}) = \{x_{n+1}\}_{n=1}^{\infty}$. Determine if $T$ is: injective, surjective, open, closed, invertible*

*Proof.*

- Let $(x_1, x_2, x_3, ...)$ and $(y_1, y_2, y_3, ...)$ be sequences in $c_0$ and $\alpha \in \mathbb{F}$. Observe that:

$$\begin{aligned} T((x_1, x_2, x_3, ...) + \alpha(y_1, y_2, y_3, ...)) &= T((x_1 + \alpha y_1, x_2 + \alpha y_2, x_3 + \alpha y_3, ...)) \\ &= (x_2 + \alpha y_2, x_3 + \alpha y_3, x_4 + \alpha y_4, ...) \\ &= (x_2, x_3, x_4, ...) + \alpha(y_2, y_3, y_4, ...) \\ &= T((x_1, x_2, x_3, ...)) + \alpha T((y_1, y_2, y_3, ...)) \end{aligned}$$

  Thus $T$ is linear.

- Consider the sequences $(1, 0, 0, ...)$ and $(2, 0, 0, ...)$. These are elements of $c_0$ which are not equal, but $T((1, 0, 0, ...)) = (0, 0, 0, ...) = T((2, 0, 0, ...))$. Therefore $T$ is not injective.
- Let $(x_1, x_2, x_3, ...) \in c_0$. Then $T((0, x_1, x_2, x_3, ...)) = (x_1, x_2, x_3, ...)$. Thus $T$ is surjective.
- Let $(x_1, x_2, x_3, ...) \in c_0$. Observe that:

$$\begin{aligned} \|T((x_1, x_2, x_3, ...))\|_{\ell^\infty} &= \|(x_2, x_3, x_4, ...)\|_{\ell^\infty} \\ &= \sup\{|x_i| \mid i = 2, 3, 4, ...\} \\ &\leqslant \sup\{|x_i| \mid i = 1, 2, 3, ...\} \\ &= \|(x_1, x_2, x_3, ...)\|_{\ell^\infty}. \end{aligned}$$

  Thus $T$ is bounded, hence continuous.

- Since $T$ is surjective, continuous, and linear, by the Open-Mapping Theorem $T$ is open.

- Since T is continuous, T has a closed graph.
- Since T is not bijective, T is not invertible.

□

**Exercise 1.6.4.** *Let $L : X \to Y$ be a linear map between Banach spaces. Suppose that the conditions $x_n \to 0$ and $Lx_n \to y$ imply that $y = 0$. Prove that $L$ is continuous.*

*Proof.* Let $\{(x_n, Tx_n)\}_{n=1}^{\infty} \subseteq \mathrm{Gr}(L)$ be a sequence such that $(x_n, Tx_n) \to (x, y) \in X \times Y$. Then $x_n \to x$ and $Tx_n \to y$. Define $u_n := x_n - x$. Then $u_n \to 0$ and $Tu_n \to y - Tx$. By hypothesis $y - Tx = 0$, giving $y = Tx$. Thus $(x, y) \in \mathrm{Gr}(T)$, which means T has a closed graph. By the Closed Graph Theorem, T is continuous. □

**Exercise 1.6.5.** *Prove that if a closed map has an inverse then the inverse is also closed.*

*Proof.* Let $\{y_n\}_{n=1}^{\infty}$ be a sequence in Y such that $y_n \to y$ and $T^{-1}y_n \to x$. Since T admits an inverse, it is bijective. Hence for each $y_n \in Y$, there exists $x_n \in X$ such that $y_n = Tx_n$. The two sequences above can be rewritten as $Tx_n \to y$ and $T^{-1}(Tx_n) = x_n \to x$. Since T has a closed graph, $y = Tx$. Since T is bijective, $x = T^{-1}y$. Thus $T^{-1}$ has a closed graph. □

**Exercise 1.6.6.** *Let $L : X \to Y$ be a continuous linear surjection, where X and Y are Banach spaces. Let $y_n \to y$ in Y. Prove that there exists points $x_n \in X$ and a constant $c \in \mathbb{R}$ such that $Lx_n = y_n$, the sequence $\{x_n\}_{n=1}^{\infty}$ converges, and $\|x_n\| \leqslant c\|y_n\|$.*

*Proof.* □

## § 1.7. Continuous Dual, Adjoint Maps, and Weak Convergence

**Definition 1.7.1.** Let X and Y be normed linear spaces and $T \in \mathcal{L}(X, Y)$. The *adjoint of* T is a map $T^* : Y^* \to X^*$ given by $T^*(\varphi) = \varphi(Tx)$.

**Proposition 1.7.1.** *Let X and Y be normed linear spaces and $T \in \mathcal{L}(X, Y)$. Then $T^* \in \mathcal{L}(Y^*, X^*)$, and $\|T^*\|_{\mathcal{L}(Y^*,X^*)} = \|T\|_{\mathcal{L}(X,Y)}$.*

*Proof.* □

### Exercises

**Exercise 1.7.1.** *Let X be a normed linear space, let $K \subset X$, and let $\Omega \subset X^*$. Show that $K^\perp$ is closed in $X^*$ and that $\Omega_\perp$ is closed in X.*

*Proof.* Let $\{\varphi_n\}_{n=1}^{\infty} \subset K^{\perp}$ be a sequence such that $\varphi_n \to \Phi \in X^*$. Let $k \in K$ be arbitrary. Then $\varphi_n(k) \to \Phi(k)$. Since $\varphi_n(k) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\Phi(k) = 0$. Since $k$ was arbitrary, we have $\Phi \in K^{\perp}$. Since $K^{\perp}$ contains all of its limit points, it is closed in $X^*$.

Let $\{x_n\}_{n=1}^{\infty} \subset \Omega_{\perp}$ be a sequence such that $x_n \to y \in X$. Let $\varphi \in \Omega$ be arbitrary. Then $\varphi(x_n) \to \varphi(y)$ since $\varphi$ is continuous. Since $\varphi(x_n) = 0$ for all $n \in \mathbb{N}$, it must be the case that $\varphi(y) = 0$. Since $\varphi$ was arbitrary, $y \in \Omega_{\perp}$. Since $\Omega_{\perp}$ contains all of its limit points, it is closed in $X$. □

**Exercise 1.7.2.** *Let $X$ be a normed linear space and $K \subset X$. Show that $(K^{\perp})_{\perp}$ is the closure of the linear space $K$ in $X$.*

*Proof.* □

## § 1.8. $L^p$ Spaces

### Exercises

**Exercise 1.8.1.** *For which $r \in \mathbb{R}$ is the function $f(x) = \frac{1}{|x|^r}$ in $L^1([0,1])$? What about $L^p([0,1])$?*

*Proof.* Since $x \in [0,1]$, we have $|x| = x$. Note that:

$$\int_0^1 x^{-r}\,dx = -\frac{1}{(r-1)x^{r-1}}\bigg|_0^1$$

The right-hand side of this equation can only be evaluated if $r-1 > 0$. Hence $\frac{1}{|x|^r} \in L^1([0,1])$ provided $r > 1$. Similarly, note that:

$$\begin{aligned}\int_0^1 (x^{-r})^p\,dx &= \int_0^1 x^{-rp}\,dx \\ &= -\frac{1}{(rp-1)x^{rp-1}}\bigg|_0^1\end{aligned}$$

The right-hand side of this equation can only be evaluation if $rp - 1 > 0$. Hence $\frac{1}{|x|^r} \in L^p([0,1])$ provided $r > \frac{1}{p}$. □

# Chapter 2
## Hilbert Spaces

### § 2.1. Inner Product Spaces

**Definition 2.1.1.** Let $V$ be a vector space over $\mathbb{F}$. An *inner product* is a map:

$$(\cdot,\cdot) : V \to \mathbb{F}$$

satisfying for all $x, y \in V$:

(1) $(x,x) \geqslant 0$ and $(x,x) = 0$ if and only if $x = 0$;

(2) $(\cdot,\cdot)$ is linear in the first slot.

(3) $(x,y) = \overline{(y,x)}$.

*Remark.* Note that an inner product over $\mathbb{R}$ (resp. $\mathbb{C}$) can be defined as a positive-definite and symmetric (resp. Hermitian) bilinear form (resp. sesquilinear form). From (2) and (3) we have:

$$\begin{aligned}(x,\lambda y) &= \overline{(\lambda y, x)}\\ &= \overline{\lambda(y,x)}\\ &= \overline{\lambda}\,\overline{(y,x)}\\ &= \overline{\lambda}(x,y).\end{aligned}$$

**Definition 2.1.2.** Let $V$ be an inner product space. We say $x, y \in V$ are *orthogonal* and write $x \perp y$ if $(x,y) = 0$.

**Example 2.1.1.**

(1) The space $\mathbb{C}^n$ with $(x,y) := \sum_{j=1}^{n} x_j \overline{y_j}$ is an inner product space.

(2) The space $\ell^2$ with $(x,y) := \sum_{j=1}^{\infty} x_j \overline{y_j}$ is an inner product space. Note that the normed induced by the inner product is precisely the familiar $\ell^2$-norm.

(3) The space $L^2(\Omega)$ with $(f,g) := \int_\Omega f(x)\overline{g(x)}dx$ is an inner product space. The norm induced by the inner product is also the familiar $L^2$-norm.

**Proposition 2.1.1.** *Let $X$ be an inner product space and $x, y \in X$, $y \neq 0$. There exists $c \in \mathbb{F}$ such that $(x - cy) \perp y$.*

*Proof.* By definition $(x - cy) \perp y$ means $(x - cy, y) = 0$. Hence $(x,y) - c(y,y) = 0$. Solving for $c$ gives $c = \frac{(x,y)}{\|y\|^2}$. ◻

**Theorem 2.1.1.** *Let $X$ be an inner product space. Let $x, y \in X$.*

*(1) If $x \perp y$, then $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ (Pythagorean Theorem).*

*(2) $|(x,y)| \leqslant \|x\| \, \|y\|$ (Cauchy-Schwartz Inequality).*

*(3) $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (Parallelogram law).*

*Proof.* (1) Observe that:

$$\begin{aligned}
\|x+y\|^2 &= (x+y, x+y) \\
&= (x,x) + (x,y) + (y,x) + (y,y) \\
&= (x,x) + (x,y) + \overline{(x,y)} + (y,y) \\
&= (x,x) + 2\Re((x,y)) + (y,y) \\
&= (x,x) + (y,y) \\
&= \|x\|^2 + \|y\|^2 .
\end{aligned}$$

(2) Assume $y \neq 0$. Find $c$ such that $(x - cy) \perp y$. Then:

$$\begin{aligned}
\|x\|^2 &= \|(x-cy) + cy\|^2 \\
&= \|(x-cy)\|^2 + \|cy\|^2 \\
&\geqslant |c|^2 \|y\|^2 .
\end{aligned}$$

Note that $c = \frac{(x,y)}{\|y\|^2}$ by Proposition 2.1.1. Hence $\|x\|^2 \geqslant \frac{|(x,y)|^2}{\|y\|^4} \|y\|^2$. Solving for $|(x,y)|$ establishes the inequality.

(3) Exercise 2.1.1. □

**Proposition 2.1.2.** *Let $V$ be an inner product space. Define $\|\cdot\| : V \to \mathbb{F}$ by $\|x\| = \sqrt{(x,x)}$. Then $(V, \|\cdot\|)$ is a normed linear space.*

*Proof.* Let $v, w \in V$ and $\alpha \in \mathbb{F}$. We have that:

$$\begin{aligned}
\|\alpha v\| &= \sqrt{(\alpha v, \alpha v)} \\
&= \sqrt{\alpha \overline{\alpha} (v,v)} \\
&= \sqrt{|\alpha|^2 (v,v)} \\
&= |\alpha| \, \|v\| .
\end{aligned}$$

Thus $\|\cdot\|$ is homogeneous. By Cauchy-Schwartz:

$$\begin{aligned}
\|v+w\|_2^2 &= \langle v+w, v+w \rangle \\
&= \langle v,v \rangle + \langle v,w \rangle + \langle w,v \rangle + \langle w,w \rangle \\
&= \|v\|_2^2 + \langle v,w \rangle + \overline{\langle v,w \rangle} + \|w\|_2^2 \\
&= \|v\|_2^2 + 2\Re(\langle v,w \rangle) + \|w\|_2^2 \\
&\leqslant \|v\|_2^2 + 2|\langle v,w \rangle| + \|w\|_2^2 \\
&\leqslant \|v\|_2^2 + 2\|v\|_2 \|w\|_2 + \|w\|_2^2 \\
&= (\|v\|_2 + \|w\|_2)^2 ,
\end{aligned}$$

Squaring both sides establishes the triangle inequality. For positive-definiteness, suppose $\|v\| = 0$. Then $\sqrt{(v, v)} = 0$, so it must be the case that $v = 0$. □

**Definition 2.1.3.** Let $X$ be an inner product space and $A \subset X$ a subspace. The *orthogonal complement of* $A$ is the set $A^\perp := \{x \in X \mid (x, a) = 0 \text{ for all } a \in A\}$.

**Proposition 2.1.3.** *If* $X$ *is an inner product space and* $A \subset X$ *is a subspace, then* $A^\perp$ *is closed in* $X$.

*Proof.* Define $\varphi_a : X \to \mathbb{F}$ by $x \mapsto (x, a)$. This map is continuous. Then $A^\perp = \bigcap_{a \in A} \varphi_a^{-1}(\{0\})$. Since $\varphi_a$ is continuous, $A^\perp$ is closed. □

## Exercises

**Exercise 2.1.1.** *Prove the Parallelogram Law in any inner-product space.*

*Proof.* Observe that:

$$\begin{aligned}
\|x+y\|^2 + \|x-y\|^2 &= (x+y, x+y) + (x-y, x-y) \\
&= (x,x) + 2\Re((x,y)) + (y,y) + (x,x) - 2\Re((x,y)) + (y,y) \\
&= 2(x,x) + 2(y,y) \\
&= 2\|x\|^2 + 2\|y\|^2 .
\end{aligned}$$

□

**Exercise 2.1.2.** *Let* $X$ *be an inner-product space over* $\mathbb{C}$ *and fix* $y \in X$. *Prove that the map* $\varphi_y : X \to \mathbb{C}$ *defined by* $\varphi_y(x) = (x, y)$ *is continuous.*

*Proof.* Let $x_1, x_2 \in X$. Observe that:

$$\begin{aligned}
|\varphi_y(x_1) - \varphi_y(x_2)| &= |(x_1, y) - (x_2, y)| \\
&= |(x_1 - x_2, y)| \\
&\leqslant \|x_1 - x_2\| \, \|y\| .
\end{aligned}$$

Since $\varphi_y$ is Lipschitz, $\varphi_y$ is continuous. □

**Exercise 2.1.3.** *Prove that in an inner-product space* $X$, *if* $\{x_n\}_{n=1}^\infty \subset X$ *is a sequence and* $y \in X$ *satisfy:*

$$\|x_n\| \to \|y\| \text{ and } (x_n, y) \to \|y\|^2 ,$$

*then* $x_n \to y$.

*Proof.* Observe that:

$$\begin{aligned}
\lim_{n\to\infty} \|x_n - y\|^2 &= \lim_{n\to\infty} (x_n - y, x_n - y) \\
&= \lim_{n\to\infty} \Big((x_n, x_n) - (y, x_n) - (x_n, y) + (y, y)\Big) \\
&= \lim_{n\to\infty} \left(\|x_n\|^2 - 2\Re((x_n, y)) + \|y\|^2\right) \\
&= \lim_{n\to\infty} \|x_n\|^2 - \lim_{n\to\infty} 2\Re((x_n, y)) + \lim_{n\to\infty} \|y\|^2 \\
&= \lim_{n\to\infty} \|x_n\|^2 - 2\Re(\lim_{n\to\infty} (x_n, y)) + \lim_{n\to\infty} \|y\|^2 \\
&= \|y\|^2 - 2\|y\|^2 + \|y\|^2 \\
&= 0.
\end{aligned}$$

Since $\|x_n - y\|^2 \to 0$, it must be the case that $\|x_n - y\| \to 0$. Thus $x_n \to y$. □

## § 2.2. Hilbert Spaces

**Definition 2.2.1.** An inner product space which is complete is called a *Hilbert space*.

*Remark.* It is standard to assume that the base field of a Hilbert space is $\mathbb{C}$.

**Theorem 2.2.1.** *Let $\mathcal{H}$ be a Hilbert space and $K \subset \mathcal{H}$ a nonempty subset which is convex and closed. For any $x \in \mathcal{H}$, there exists a unique $y \in K$ which is closed to $x$.*

*Proof.* Let $x \in X$. Let $\alpha := \operatorname{dist}(x, K)$. Find a sequence $\{y_n\}_{n=1}^{\infty} \subset K$ such that $\|x - y_n\| \to \alpha$ as $n \to \infty$. For any $n, m \in \mathbb{N}$:

$$\begin{aligned}
\|y_n - y_m\|^2 &= \|(y_n - x) + (x - y_m)\|^2 \\
&= 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - \|y_n + y_m - 2x\|^2 && \text{Parallelogram Law} \\
&= 2\|y_n - x\|^2 + 2\|x - y_m\|^2 - 4\left\|\frac{y_n + y_m}{2} - x\right\|^2 && \text{See footnote[1]} \\
&\leqslant 2\|y_n - x\|^2 + 2\|y_m - x\|^2 - 4\alpha^2.
\end{aligned}$$

Since $\|x - y_n\| \to \alpha^2$ and $\|x - y_m\| \to \alpha^2$, note that $\|y_n - y_m\|^2$ can be made arbitrary small. Hence $\{y_n\}_{n=1}^{\infty}$ is Cauchy. Since $K$ is a closed subset of a Hilbert space, it is complete. Hence there exists some $y \in K$ such that $y_n \to y$. Finally:

$$\begin{aligned}
\|x - y\| &= \left\|x - \lim_{n\to\infty} y_n\right\| \\
&= \lim_{n\to\infty} \|x - y_n\| \\
&= \alpha.
\end{aligned}$$

[1] Note $\frac{y_n + y_m}{2} \in K$ because $K$ is convex.

Now if $y'$ is any other element in $K$ with $\|y' - x\| = \alpha$, by our previous calculation:

$$\begin{aligned}\|y - y'\|^2 &= 2\|y - x\|^2 + 2\|y' - x\|^2 - \|y + y' - 2x\|^2 \\ &\leqslant 2\alpha^2 + 2\alpha^2 - 4\alpha^2 \\ &= 0.\end{aligned}$$

Thus $y = y'$. ◻

**Theorem 2.2.2.** *Let $\mathcal{H}$ be a Hilbert space. Let $Z \subset H$ be a closed subspace. Then $\mathcal{H} = Z \oplus Z^\perp$.*

*Proof.* If $z \in Z \cap Z^\perp$, then $z \perp z$; i.e., $(z, z) = 0$. Therefore $z = 0$.

Now let $x \in \mathcal{H}$. By Theorem 2.2.1, let $z \in Z$ be the unique element closet to $x$. We show that $x - z \in Z^\perp$. Let $z' \in Z$ be nonzero. By Proposition 2.1.1, choose $\lambda \in \mathbb{C}$ such that $\lambda z' \perp (x - z) - \lambda z'$. Write:

$$x - z = \lambda z' + (x - z) - \lambda z'.$$

Then the Pythagorean Theorem gives:

$$\begin{aligned}\|x - z\|^2 &= \|\lambda z'\|^2 + \|x - (z + \lambda z')\|^2 \\ &\geqslant \|\lambda z'\|^2 + \|x - z\|^2.\end{aligned}$$

Note that the inequality is a result from the fact that $z$ is the closest element in $Z$ to $x$. Subtracting $\|x - z\|^2$ from both sides yields:

$$0 \geqslant \|\lambda z'\|^2$$

Because $z' \neq 0$, this forces $\lambda = 0$. Since $\lambda = \frac{(x-z,z')}{\|z'\|^2}$, we obtain $(x - z, z') = 0$. As $z' \in Z$ was arbitrary, $x - z \in Z^\perp$. Consequently, $x = z + (x - z) \in Z + Z^\perp$, establishing $x \in Z \oplus Z^\perp$. ◻

**Definition 2.2.2.** Let $X$ be an inner product space. Let $\mathcal{U} := \{u_\alpha \mid \alpha \in I\} \subset X$. We say $\mathcal{U}$ is <u>orthogonal</u> if for all $\alpha \neq \beta$, we have $(u_\alpha, u_\beta) = 0$. We say $\mathcal{U}$ is <u>orthonormal</u> if for all $\alpha, \beta \in I$, we have:

$$(u_\alpha, u_\beta) = \begin{cases} 1, & \alpha = \beta \\ 0, & \alpha \neq \beta. \end{cases}$$

**Theorem 2.2.3.** *Let $X$ be a nonzero inner product space. There exists a maximal orthonormal set by inclusion*

*Proof.* Order all possible orthonormal sets by inclusion. Show that every chain admits an upper bound. Then apply Zorn's Lemma. ◻

**Theorem 2.2.4.** *Let $X$ be a separable inner product space. Then any orthonormal subset is at most countable.*

*Proof.* Let $\{a_j\}_{j=1}^{\infty}$ be a countable dense subset of $X$. Suppose towards contradiction $\mathcal{U} := \{u_\alpha \mid \alpha \in I\}$ is an uncountable orthonormal set in $X$. Pick $\alpha, \beta \in I$ such that $\alpha \neq \beta$. Then:

$$\begin{aligned}\left\|u_\alpha - u_\beta\right\|^2 &= \|u_\alpha\|^2 + \left\|u_\beta\right\|^2 \\ &= 2.\end{aligned}$$

Therefore, any two elements of $\mathcal{U}$ are $\sqrt{2}$ apart. Now for all $\alpha \in I$, consider the open balls $B(u_\alpha, \frac{\sqrt{2}}{2})$. Since each open ball has diameter less than $\sqrt{2}$, each ball can contain at most one element of $\{a_j\}_{j=1}^{\infty}$. Moreover, the collection $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$ is disjoint. Indeed, if $y \in B(u_\alpha, \frac{\sqrt{2}}{2}) \cap B(u_\beta, \frac{\sqrt{2}}{2})$, then $\left\|u_\alpha - u_\beta\right\| \leqslant \|u_\alpha - y\| + \left\|y - u_\beta\right\| < \sqrt{2}$, which is a contradiction. Since $\{a_j\}_{j=1}^{\infty}$ is a dense subset, each $a_j$ must be contained in a ball. We can define a surjection from $\{a_j\}_{j=1}^{\infty}$ to $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$. But since $\{B(u_\alpha, \frac{\sqrt{2}}{2}) \mid \alpha \in I\}$ is indexed by an uncountable set, it must be the case that $\{a_j\}_{j=1}^{\infty}$ is uncountable. This is a contradiction, whence any orthonormal subset must be at most countable. □

**Theorem 2.2.5.** *Let* $X$ *be an inner product space. Let* $Y \subset X$ *be a subspace with* $\dim Y = n$.

*(1)* $Y$ *has an orthonormal basis* $\{u_1, ..., u_n\}$;

*(2) For each* $y \in Y$, *we can write* $y = \sum_{j=1}^{n}(y, u_j)u_j$. *Moreover,* $\|y\|^2 = \sum_{j=1}^{n} |(y, u_j)|^2$.

*(3) There is an isometric isomorphism* $Y \to \mathbb{C}^n$.

*(4) For any* $x \in X$, *we have* $x - \sum_{j=1}^{n}(x, u_j)u_j \perp Y$. *Moreover,* $\sum_{j=1}^{n}(x, u_j)u_j$ *is the unique element in* $Y$ *closest to* $x$,

*Proof.* (1) Apply the Gram-Schmidt process to a basis of $Y$.

(2) Let $y \in Y$. We can uniquely write $y = \sum_{j=1}^{n} a_j u_j$ for some $a_1, ..., a_n \in \mathbb{C}$. Then $(y, u_j) = \left(\sum_{k=1}^{n} a_k u_k, u_j\right) = \sum_{j=1}^{n} a_k(u_k, u_j) = a_j$. Hence $y = \sum_{j=1}^{n}(y, u_j)u_j$.

(3) For any $y \in Y$, we can write $y = \sum_{j=1}^{n} a_j u_j$. Then $Y \to \mathbb{C}^n$ given by $y \mapsto (a_1, ..., a_n)$ is linear, bijective, and norm preserving.

(4) Let $x \in X$. Define $a_j := (x, u_j)$ for each $j = 1, ..., n$. For each $k = 1, ..., n$:

$$\begin{aligned}\left(x - \sum_{j=1}^{n} a_j u_j, u_k\right) &= (x, u_k) - \left(\sum_{j=1}^{n} a_j u_j, u_k\right) \\ &= (x, u_k) - \sum_{j=1}^{n} a_j(u_j, u_k) \\ &= (x, u_k) - a_k \\ &= (x, u_k) - (x, u_k) \\ &= 0.\end{aligned}$$

Since any $y \in Y$ can be uniquely written as a linear combination of $u_1, ..., u_n$, the above string of equalities implies $x - \sum_{j=1}^{n}(x, u_j)u_j \perp Y$. Now let $\sum_{j=1}^{n} b_j u_j$ be any element in $Y$.

Then:

$$\begin{aligned}\left\|x - \sum_{j=1}^{n} b_j u_j\right\|^2 &= \left\|x - \sum_{j=1}^{n} a_j u_j + \sum_{j=1}^{n} (a_j - b_j) u_j\right\|^2 \\ &= \left\|x - \sum_{j=1}^{n} a_j u_j\right\|^2 + \left\|\sum_{j=1}^{n} (a_j - b_j) u_j\right\|^2 \\ &= \left\|x - \sum_{j=1}^{n} a_j u_j\right\|^2 + \sum_{j=1}^{n} |a_j - b_j|^2 \\ &\geqslant \left\|x - \sum_{j=1}^{n} a_j u_j\right\|^2,\end{aligned}$$

where we used the fact that $x - \sum_{j=1}^{n} a_j u_j \perp \sum_{j=1}^{n} (a_j - b_j) u_j$. This establishes that $\sum_{j=1}^{n} a_j u_j$ is the unique element in $Y$ closest to $x$, where we have equality only when $a_j = b_j$ for all $j = 1, ..., n$. □

**Theorem 2.2.6.** *Let $\mathcal{H}$ be a Hilbert space. Let $\{x_j\}_{j=1}^{\infty}$ be an orthogonal sequence in $\mathcal{H}$. Then $\sum_{j=1}^{\infty} x_j$ converges in $\mathcal{H}$ if and only if $\sum_{j=1}^{\infty} \|x_j\|^2 < \infty$.*

*Proof.* Define:

$$\begin{aligned} z_n &:= \sum_{j=1}^{n} x_j, \\ s_n &:= \sum_{j=1}^{n} \|x_j\|^2. \end{aligned}$$

Now $\sum_{j=1}^{\infty} x_j$ converges if and only if $\{z_n\}_{n=1}^{\infty}$ is Cauchy. For $n > m$:

$$\begin{aligned}\|z_n - z_m\|^2 &= \left\|\sum_{j=m+1}^{n} x_j\right\|^2 \\ &= \sum_{j=m+1}^{n} \|x_j\|^2 \\ &= |s_n - s_m|.\end{aligned}$$

From this, $\{z_n\}_{n=1}^{\infty}$ is Cauchy in $\mathcal{H}$ if and only if $\{s_n\}_{n=1}^{\infty}$ is Cauchy in $\mathbb{C}$. □

**Theorem 2.2.7** (Bessel's Inequality)**.** *Let $X$ be an inner-product space. Let $\{u_j\}_{j=1}^{\infty}$ be an orthonormal sequence in $X$. For all $x \in X$, we have $\sum_{j=1}^{\infty} |(x, u_j)|^2 \leqslant \|x\|^2$.*

*Proof.* For each $n \in \mathbb{N}$, define $Y_n := \operatorname{span}\{u_1, ..., u_n\}$. Let $P_{Y_n}$ denote the projection onto $Y_n$. Then:

$$\begin{aligned}\sum_{j=1}^{n} |(x, u_j)|^2 &= \sum_{j=1}^{n} \left\|(x, u_j)u_j\right\|^2 \\ &= \left\|\sum_{j=1}^{n} (x, u_j)u_j\right\|^2 \\ &= \|P_{Y_n}(x)\|^2 \\ &\leqslant \|P_{Y_n}\|^2 \|x\|^2 \\ &= \|x\|^2 .\end{aligned}$$

Note that $\|P_{Y_n}\| = 1$ is established in Exercise 2.2.5. Since $n$ was arbitrary, by the Monotone Convergence Theorem $\sum_{j=1}^{\infty} |(x, u_j)|^2 \leqslant \|x\|^2$. □

**Theorem 2.2.8.** *Let $\mathcal{H}$ be a Hilbert space. Let $\{u_j\}_{j=1}^{\infty}$ be an orthonormal sequence in $H$. The following are equivalent:*

*(1) $\{u_j\}_{j=1}^{\infty}$ is a maximal orthonormal set;*

*(2) For all $x \in \mathcal{H}$, if $x \perp \{u_j\}_{j=1}^{\infty}$, then $x = 0$;*

*(3) For all $x \in \mathcal{H}$, there exists a unique sequence $\{a_j\}_{j=1}^{\infty}$ such that $x = \sum_{j=1}^{\infty} a_j u_j$.*

*Proof.* We will only show (2) if and only if (3), as (1) if and only if (2) follows immediately by applying definitions.

Suppose for all $x \in \mathcal{H}$ we have $x \perp \{u_j\}_{j=1}^{\infty}$ implies $x = 0$. Let $x \in \mathcal{H}$. Claim: $x - \sum_{j=1}^{\infty}(x, u_j)u_j \perp \{u_j\}_{j=1}^{\infty}$. Indeed, for any $u_k \in \{u_j\}_{j=1}^{\infty}$:

$$\begin{aligned}\left(x - \sum_{j=1}^{\infty}(x, u_j)u_j, u_k\right) &= \lim_{n\to\infty}\left(x - \sum_{j=1}^{n}(x, u_j)u_j, u_k\right) \\ &= \lim_{n\to\infty}\left((x, u_k) - \sum_{j=1}^{n}\big((x, u_j)u_j, u_k\big)\right) \\ &= \lim_{n\to\infty}\big((x, u_k) - (x, u_k)\big) \\ &= 0.\end{aligned}$$

Hence $x - \sum_{j=1}^{\infty}(x, u_j)u_j = 0$, and rearranging gives $x = \sum_{j=1}^{\infty}(x, u_j)u_j$. Now suppose there exists a sequence $\{b_n\}_{n=1}^{\infty}$ such that $x = \sum_{j=1}^{\infty} b_j u_j$. Then $0 = \sum_{j=1}^{\infty}\big((x, u_j) - b_j\big)u_j$, which by Bessel's Inequality results in $\sum_{j=1}^{\infty} |(x, u_j) - b_j|^2 \leqslant 0$. It must be the case that $(x, u_j) = b_j$ for all $j \in \mathbb{N}$, establishing uniqueness.

Let $x \in X$ and suppose $x \perp \{u_j\}_{j=1}^{\infty}$. Write $x = \sum_{j=1}^{\infty}(x, u_j)u_j$. Then for all $k \in \mathbb{N}$:

$$\begin{aligned}(x, u_k) &= \left(\sum_{j=1}^{\infty}(x, u_j)u_j, u_k\right) \\ &= 0.\end{aligned}$$

Thus $x = 0$. $\square$

**Definition 2.2.3.** Let $\mathcal{H}$ be a Hilbert space and $\{u_j\}_{j=1}^{\infty}$ an orthonormal sequence in $\mathcal{H}$. If any of the three equivalent statements in Theorem 2.2.8 hold true, we say $\{u_j\}_{j=1}^{\infty}$ is a *orthonormal basis* (or *Hilbert basis*).

**Definition 2.2.4.** Let $\mathcal{H}$ be a Hilbert space and $\{u_j\}_{j=1}^{\infty}$ an orthonormal basis. Let $x \in \mathcal{H}$.

(1) The sequence $\{(x, u_j)\}_{j=1}^{\infty}$ is referred to as the *generalized Fourier coefficients of* $x$.

(2) The series $\sum_{j=1}^{\infty}(x, u_j)u_j$ is called the *generalized Fourier series of* $x$.

**Example 2.2.1.** The set $\{e^n\}_{n=1}^{\infty}$ is an orthonormal basis in $\ell^2$. For any separable infinite dimensional hilbert space, let $\{u_j\}_{n=1}^{\infty}$ be an orthonormal basis. Define $H \to \ell^2$ by $\sum_{j=1}^{\infty} a_j, u_j \mapsto (a_1, a_2, ...)$. This is an isometric isomorphism.

Feb 12, stuff on $L^2$

**Theorem 2.2.9** (Riez Representation Theorem)**.** *Let $\mathcal{H}$ be a Hilbert space. Let $\varphi \in \mathcal{H}^*$. There exists a unique $y \in H$ such that for all $x \in \mathcal{H}$, $\varphi(x) = (x, y)$.*

*Proof.* If $y, y' \in \mathcal{H}$ satisfy $\varphi(x) = (x, y) = (x, y')$ for all $x \in \mathcal{H}$, then $(x, y - y') = 0$ for all $x \in \mathcal{H}$. Note that $y - y' \in \mathcal{H}$, hence we must have that $(y - y', y - y') = 0$. Thus $\|y - y'\| = 0$, hence $y = y'$.

We now prove existence. Let $\varphi \in \mathcal{H}^*$. Let $z \in N(\varphi)^{\perp}$ such that $\|z\| = 1$. For any $x \in H$, we have $x - \frac{\varphi(x)}{\varphi(z)}z \in N(\varphi)$ and the equation:

$$x = \left(x - \frac{\varphi(x)}{\varphi(z)}z\right) + \left(\frac{\varphi(x)}{\varphi(z)}z\right).$$

Now:

$$\begin{aligned}(x, \lambda z) &= \left(x - \frac{\varphi(x)}{\varphi(z)}z + \frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) \\ &= \left(x - \frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) + \left(\frac{\varphi(x)}{\varphi(z)}z, \lambda z\right).\end{aligned}$$

Because $z \in N(\varphi)^{\perp}$ and $x - \frac{\varphi(x)}{\varphi(z)}z \in N(\varphi)$, then $\left(x - \frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) = 0$. We now have:

$$\begin{aligned}(x, \lambda z) &= \left(x - \frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) + \left(\frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) \\ &= \left(\frac{\varphi(x)}{\varphi(z)}z, \lambda z\right) \\ &= \frac{\varphi(x)}{\varphi(z)}\overline{\lambda}(z, z) \\ &= \frac{\varphi(x)}{\varphi(z)}\overline{\lambda}\|z\|^2 \\ &= \frac{\varphi(x)}{\varphi(z)}\overline{\lambda}.\end{aligned}$$

Let $\lambda := \overline{\varphi(z)}$. Then $(x, \varphi(z)) = \varphi(x)$, establishing the theorem. $\square$

self-adjoint operators

## Exercises

**Exercise 2.2.1.** *Let* $H = L^2([0,1])$. *Use the Gram-Schmidt orthogonalization process from linear algebra to produce an orthonormal set* $\{u_1, u_2, u_3\}$ *which spans the same space as* $\operatorname{span}\{1, x, x^2\}$.

*Proof.* Let $v_1 := 1$, $v_2 := x$, and $v_3 := x^2$. The Gram-Schmidt process gives:

$$r_1 = v_1 = 1,$$
$$\|r_1\| = \sqrt{(r_1, r_1)} = 1.$$

$$r_2 = v_2 - \frac{(v_2, r_1)}{(r_1, r_1)} r_1 = x - \frac{1}{2}.$$
$$\|r_2\| = \sqrt{(r_2, r_2)} = \sqrt{\int_0^1 \left(x - \frac{1}{2}\right)^2 dx} = \frac{1}{\sqrt{12}}$$

$$r_3 = v_3 - \frac{(v_3, r_1)}{(r_1, r_1)} r_1 - \frac{(v_3, r_2)}{(r_2, r_2)} r_2 = x^2 - x + \frac{1}{6},$$
$$\|r_3\| = \sqrt{(r_3, r_3)} = \sqrt{\int_0^1 \left(x^2 - x + \frac{1}{6}\right)^2 dx} = \frac{1}{\sqrt{180}}.$$

Thus an orthonormal basis for $\operatorname{span}\{1, x, x^2\}$ is:

$$u_1 := \frac{r_1}{\|r_1\|} = 1$$
$$u_2 := \frac{r_2}{\|r_2\|} = \sqrt{12}\left(x - \frac{1}{2}\right)$$
$$u_3 := \frac{r_3}{\|r_3\|} = \sqrt{180}\left(x^2 - x + \frac{1}{6}\right).$$

$\square$

**Exercise 2.2.2.** *Show directly that the functions:*

$$u_n(x) = e^{i2\pi n x}, \, n \in \mathbb{Z}$$

*form an orthonormal sequence in* $L^1([0,1])$. *Then write the function* $f(x) = x$ *in terms of this basis, that is find* $a_n$ *such that:*

$$x \sim \sum_{n=-\infty}^{\infty} a_n u_n(x).$$

*What does Parseval's identity say in this case? (After some algebra you should be able to use your summation to find a formula for* $\sum_{n=1}^{\infty} \frac{1}{n^2}$.*)*

*Proof.* Observe that:

$$\int_0^1 e^{i2\pi nx}e^{-i2\pi nx}\,dx = \int_0^1 1\,dx = 1.$$

$$\begin{aligned}
\int_0^1 e^{i2\pi nx}e^{-i2\pi mx}\,dx &= \int_0^1 e^{i2\pi(n-m)x}\,dx \\
&= \left.\frac{e^{i2\pi(n-m)x}}{i2\pi(n-m)}\right|_0^1 \\
&= \frac{e^{i2\pi(n-m)}-1}{i2\pi(n-m)} \\
&= \frac{\cos\big(2\pi(n-m)\big) - i\sin\big(2\pi(n-m)\big) - 1}{i2\pi(n-m)} \\
&= \frac{1-0-1}{i2\pi(n-m)} \\
&= 0.
\end{aligned}$$

Thus $\{u_n\}_{n=1}^{\infty}$ is an orthonormal sequence in $L^1([0,1])$. We proceed by cases on computing the Fourier coefficients of $x$. For $n = 0$:

$$a_0 = \int_0^1 x\,dx = \frac{1}{2}.$$

For $n \neq 0$:

$$\begin{aligned}
a_n &= (x, u_n) \\
&= \int_0^1 xe^{-i2\pi nx}\,dx \\
&= -\frac{e^{i2\pi n}(e^{i2\pi n} - i2\pi n - 1)}{4n^2\pi^2} && \text{From Mathematica} \\
&= \frac{i}{2n\pi}. && \text{Also from Mathematica}
\end{aligned}$$

Using Parseval's Identity:

$$\begin{aligned}
\frac{1}{3} &= \int_0^1 x^2 dx \\
&= \|x\|^2 \\
&= \sum_{n=-\infty}^{\infty} |a_n|^2 \\
&= |a_0|^2 + 2\sum_{n=1}^{\infty} |a_n|^2 \\
&= \frac{1}{4} + 2\sum_{n=1}^{\infty} \left|\frac{i}{2\pi n}\right|^2 \\
&= \frac{1}{4} + \frac{1}{2\pi^2}\sum_{n=1}^{\infty} \frac{1}{n^2}.
\end{aligned}$$

Solving for $\sum_{n=1}^{\infty} \frac{1}{n^2}$ gives:

$$\begin{aligned}
\sum_{n=1}^{\infty} \frac{1}{n^2} &= 2\pi^2\left(\frac{1}{3} - \frac{1}{4}\right) \\
&= \frac{2\pi^2}{12} \\
&= \frac{\pi^2}{6}.
\end{aligned}$$

□

**Exercise 2.2.3.** *Show directly that the functions:*

$$1, \cos x, \sin x, \cos 2x, \sin 2x, \cos 3x, \ldots$$

*are mutually orthogonal in* $L^2([-\pi, \pi])$.

*Proof.* For $n, m \neq 0$:

$$\begin{aligned}
\int_{-\pi}^{\pi} \cos(nx)dx &= \frac{1}{n}\sin(nx)\bigg|_{-\pi}^{\pi} = \frac{1}{n}\sin(n\pi) - \frac{1}{n}\sin(-n\pi) = \frac{2}{n}\sin(n\pi) = 0, \\
\int_{-\pi}^{\pi} \sin(nx)dx &= -\frac{1}{n}\cos(nx)\bigg|_{-\pi}^{\pi} = -\frac{1}{n}\cos(nx) + \frac{1}{n}\cos(-n\pi) = 0, \\
\int_{-\pi}^{\pi} \cos(nx)\cos(mx)dx &= \int_{-\pi}^{\pi} \frac{\cos((n+m)x) + \cos((n-m)x)}{2}dx = 0, \\
\int_{-\pi}^{\pi} \sin(nx)\sin(mx)dx &= \int_{-\pi}^{\pi} \frac{\cos((n-m)x) - \cos((n+m)x)}{2}dx = 0, \\
\int_{-\pi}^{\pi} \cos(nx)\sin(mx)dx &= \int_{-\pi}^{\pi} \frac{\sin((n+m)x) + \sin((n-m)x)}{2}dx = 0.
\end{aligned}$$

□

**Exercise 2.2.4.** *Let $\{u_n\}_{n=1}^{\infty}$ be an orthonormal basis for a Hilbert space $H$. Let $\{v_n\}_{n=1}^{\infty}$ be an orthonormal seq which satisfies $\sum_{n=1}^{\infty} \|u_n - v_n\|^2 < 1$. Show that $\{v_n\}_{n=1}^{\infty}$ is also a Hilbert basis for $H$.*

*Proof.* Let $x \in \mathcal{H}$. Suppose $x \perp \{v_j\}_{j=1}^{\infty}$. By Parseval's Identity:

$$\begin{aligned}
\|x\|^2 &= \sum_{j=1}^{\infty} |(x, u_n)|^2 \\
&= \sum_{j=1}^{\infty} |(x, u_n) - (x, v_n)|^2 \\
&= \sum_{j=1}^{\infty} |(x, u_n - v_n)|^2 \\
&\leqslant \sum_{j=1}^{\infty} \|x\|^2 \|u_n - v_n\|^2 .
\end{aligned}$$

Suppose towards contradiction $x \neq 0$. We can divide both sides of the above equation by $\|x\|^2$, giving:

$$1 \leqslant \sum_{j=1}^{\infty} \|u_n - v_n\|^2 < 1.$$

This is a contradiction, hence $x = 0$. Since we've shown $x \perp \{v_j\}_{j=1}^{\infty}$ implies $x = 0$, we've established $\{v_j\}_{j=1}^{\infty}$ as an orthonormal basis for $\mathcal{H}$. □

**Exercise 2.2.5.** *Let $H$ be a Hilbert space and $Y \neq \{0\}$ be a closed subspace, and let $P$ denote orthogonal projection onto $Y$. Show that:*

*(1) $P$ is linear;*

*(2) $P$ is bounded and $\|P\| = 1$;*

*(3) For all $x \in H$ we have $(Px, x) = \|Px\|^2$;*

*(4) $I - P$ is orthogonal projection onto $Y^{\perp}$.*

*Proof.* (a) For any $x_1, x_2 \in H$ and $\alpha \in \mathbb{C}$ we have:

$$\begin{aligned}
P(x_1 + \alpha x_2) &= P(y_1 + y_1^{\perp} + \alpha y_2 + \alpha y_2^{\perp}) \\
&= P(y_1 + \alpha y_2 + (y_1 + \alpha y_2)^{\perp}) \\
&= y_1 + \alpha y_2 \\
&= P(x_1) + \alpha P(x_2).
\end{aligned}$$

Thus $T$ is linear.

(b) Let $x \in H$. Observe that:

$$\begin{aligned}\|Px\|^2 &= \|y\|^2 \\ &\leqslant \|y\|^2 + \left\|y^\perp\right\|^2 \\ &= \left\|y + y^\perp\right\|^2 \\ &= \|x\|^2 .\end{aligned}$$

Squaring both sides gives $\|Px\| \leqslant \|x\|$. Diving by $\|x\|$ on both sides of this inequality and taking the supremum over all $x \in H$, $x \neq 0$ gives $\|P\|_{\mathcal{L}(H,Y)} \leqslant 1$. Now let $y \in B_Y(0,1)$. Then:

$$\|Py\| = \|y\| = 1.$$

Hence $1 \in \{\|Px\| \mid x \in B_H(0,1)\}$. It must be the case that $1 \leqslant \|P\|_{\mathcal{L}(H,Y)}$. Therefore $\|P\|_{\mathcal{L}(H,Y)} = 1$.

(c) We will first show that $P$ is self-adjoint and idempotent. Let $x_1, x_2 \in H$. We have:

$$\begin{aligned}(Px_1, x_2) &= (y, y_2 + y_2^\perp) \\ &= (y_1, y_2) + (y_1, y_2^\perp) \\ &= (y_1, y_2).\end{aligned}$$

$$\begin{aligned}(x_1, Px_2) &= (y_1 + y_1^\perp, y_2) \\ &= (y_1, y_2) + (y_1^\perp, y_2) \\ &= (y_1, y_2).\end{aligned}$$

Therefore $P^* = P$; i.e., $P$ is self-adjoint. Now for any $x \in H$, notice that:

$$\begin{aligned}P(P(x)) &= P(y) \\ &= y \\ &= P(x).\end{aligned}$$

Thus $P^2 = P$. Using these two facts:

$$\begin{aligned}\|Px\|^2 &= (Px, Px) \\ &= (P^*Px, x) \\ &= (P^2x, x) \\ &= (Px, x).\end{aligned}$$

□

(d) For any $x \in H$:

$$\begin{aligned}(I - P)(x) &= Ix - Px \\ &= y + y^\perp - P(y + y^\perp) \\ &= y + y^\perp - y \\ &= y^\perp\end{aligned}$$

**Exercise 2.2.6.** *Let* $H$ *be a Hilbert Space with orthonormal basis* $\{u_n\}_{n=1}^{\infty}$. *We try to define a map:*

$$T : H \to H \text{ by } Tx = \sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n$$

*for a sequence* $(\lambda_n) \subset \mathbb{C}$. *First show that this map is a bounded linear operator if and only if* $(\lambda_n)$ *is a bounded sequence. Then, under what condition is* $T$ *self-adjoint?*

*Proof.* Suppose $T \in \mathcal{L}(H)$. Then:

$$\begin{aligned} |\lambda_n| &= |\lambda_n| \, \|u_n\| \\ &= \|\lambda_n u_n\| \\ &= \|Tu_n\| \\ &\leqslant \|T\| \, \|u_n\| \\ &= \|T\| \, . \end{aligned}$$

Taking the supremum over all $n \in \mathbb{N}$ gives $\sup_{n\in\mathbb{N}} |\lambda_n| \leqslant \|T\|_{\mathcal{L}(H)} < \infty$. Therefore $(\lambda_n)$ is bounded. Conversely, suppose $\sup_{n\in\mathbb{N}} |\lambda_n| := M \leqslant \infty$. We must first show that $T$ is linear. Let $x, y \in H$ and $\alpha \in \mathbb{C}$. Define $a_n := (x, u_n)$ and $b_n := (y, u_n)$. We can write:

$$\begin{aligned} x + \alpha y &= \sum_{n=1}^{\infty} a_n u_n + \alpha \sum_{n=1}^{\infty} b_n u_n \\ &= \sum_{n=1}^{\infty} (a_n + \alpha b_n) u_n. \end{aligned}$$

Applying $T$ to $x + \alpha y$ gives:

$$\begin{aligned} T(x + \alpha y) &= \sum_{n=1}^{\infty} \lambda_n (a_n + \alpha b_n) u_n \\ &= \sum_{n=1}^{\infty} \lambda_n a_n u_n + \alpha \sum_{n=1}^{\infty} \lambda_n b_n u_n \\ &= Tx + \alpha Ty. \end{aligned}$$

Thus T is linear. Now let $x \in X$ and again define $a_n := (x, u_n)$. By Parseval's identity:

$$\begin{aligned}
\|Tx\|^2 &= \sum_{n=1}^{\infty} |(Tx, u_n)|^2 \\
&= \sum_{n=1}^{\infty} \left| \left( \sum_{j=1}^{\infty} \lambda_j a_n u_j, u_n \right) \right|^2 \\
&= \sum_{n=1}^{\infty} \left| \sum_{j=1}^{\infty} \left( \lambda_j a_j u_j, u_n \right) \right|^2 \\
&= \sum_{n=1}^{\infty} |\lambda_n a_n|^2 \\
&\leqslant \sum_{n=1}^{\infty} |M|^2 |a_n|^2 \\
&= |M|^2 \sum_{n=1}^{\infty} |a_n|^2 \\
&= |M|^2 \|x\|^2 .
\end{aligned}$$

Rearranging and taking the square-root of both sides yields:

$$\frac{\|Tx\|}{\|x\|} \leqslant |M|.$$

Taking the supremum over all $x \in X$, $x \neq 0$ gives $\|T\|_{\mathcal{L}(H)} \leqslant |M| < \infty$. Thus $T \in \mathcal{L}(H)$.

Let $x, y \in H$. Observe that:

$$\begin{aligned}
(Tx, y) &= \left( \sum_{n=1}^{\infty} \lambda_n (x, u_n) u_n, y \right) \\
&= \sum_{n=1}^{\infty} \left( \lambda_n (x, u_n) u_n, y \right) \\
&= \sum_{n=1}^{\infty} \lambda_n (x, u_n)(u_n, y).
\end{aligned}$$

$$\begin{aligned}
(x, Ty) &= \left( x, \sum_{n=1}^{\infty} \lambda_n (y, u_n) u_n \right) \\
&= \sum_{n=1}^{\infty} \left( x, \lambda_n (y, u_n) u_n \right) \\
&= \sum_{n=1}^{\infty} \overline{\lambda_n (y, u_n)} (x, u_n) \\
&= \sum_{n=1}^{\infty} \overline{\lambda_n} (x, u_n)(y, u_n).
\end{aligned}$$

From this, $T$ is self-adjoint if and only if $\lambda_n = \overline{\lambda_n}$; i.e., whenever $(\lambda_n) \subset \mathbb{R}$. $\square$

## § 2.3. Compact Operators

**Definition 2.3.1.** Let $X$ and $Y$ be normed linear spaces and $T : X \to Y$ linear. We say $T$ is *compact* if for every bounded sequence $\{x_n\}_{n=1}^{\infty} \subset X$, the sequence $\{Tx_n\}_{n=1}^{\infty}$ has a convergent subsequence. Equivalently, the operator $T$ is said to be *compact* if $\overline{T(B_X(0,1))}$ is compact in $Y$.

**Proposition 2.3.1.** *Let $X$ and $Y$ be normed linear spaces and $T : X \to Y$ linear.*

*(1) If $T$ is compact, then $T$ is bounded.*

*(2) If $T \in \mathcal{L}(X, Y)$ and $R(T)$ is finite-dimensional, then $T$ is compact.*

*Proof.* (1) Let $x \in X$, $\|x\| = 1$. Then $Tx \in T(B_X(0,1)) \subseteq \overline{T(B_X(0,1))}$. Since $T$ is compact, $\overline{T(B_X(0,1))}$ is compact in $Y$. Since $\overline{T(B_X(0,1))}$ is compact, it is bounded. Hence $\|Tx\| < \infty$.

(2) Since $T \in \mathcal{L}(X, Y)$, the subspace $T(B_X(0,1))$ is bounded. Since $R(T)$ is finite-dimensional, by Corollary 1.2.2 $T(B_X(0,1))$ is closed. Hence $T(B_X(0,1)) = \overline{T(B_X(0,1))}$ is closed and bounded. By Theorem 1.2.1, $\overline{T(B_X(0,1))}$ is compact. Thus $T$ is compact. $\square$

**Example 2.3.1.** Let $X$ be a normed linear space. Then $\mathrm{id} : X \to X$ is bounded, but not compact.

Define $T : \ell^2 \to \ell^2$ by $e^n \mapsto \frac{1}{\sqrt{n}} e^n$ and extend linearly. This map is compact, but $R(T)$ is not finite dimensional.

**Theorem 2.3.1.** *Let $X$ be a normed linear space and $Y$ a Banach space. The set:*

$$\{T \in \mathcal{L}(X, Y) \mid T \text{ is compact } \}$$

*is closed in $\mathcal{L}(X, Y)$.*

*Proof.* Let $\{T_n\}_{n=1}^{\infty}$ be a sequence of compact operators converging to $T \in \mathcal{L}(X, Y)$. Let $\{x_n\}_{n=1}^{\infty}$ be a bounded sequence. Choose a subsequence $\{x_{1,j}\}_{j=1}^{\infty} \subset \{x_n\}_{n=1}^{\infty}$ such that $\{T_1(x_{1,j})\}_{j=1}^{\infty}$ is convergent in $Y$. Note that $\{x_{1,j}\}_{j=1}^{\infty}$ is still bounded. Choose a subsequence $\{x_{2,j}\}_{j=1}^{\infty} \subset \{x_{1,j}\}_{j=1}^{\infty} \subset \{x_n\}_{n=1}^{\infty}$ such that $\{T_2(x_{2,j})\}_{j=1}^{\infty}$ is convergent in $Y$. In general, choose $\{x_{k,j}\}_{j=1}^{\infty} \subset \{x_{k-1,j}\}_{j=1}^{\infty} \subset \ldots \subset \{x_n\}_{n=1}^{\infty}$ such that $\{T_k(x_{k,j})\}_{j=1}^{\infty}$ is convergent. Define $\{y_n\}_{n=1}^{\infty}$ by $y_n = x_{n,n}$. Then $\{y_n\}_{n=1}^{\infty} \subset \{x_n\}_{n=1}^{\infty}$. Then for any $k < n$, we have that $\{T_k y_n\}_{n=1}^{\infty}$ is convergent. $\square$